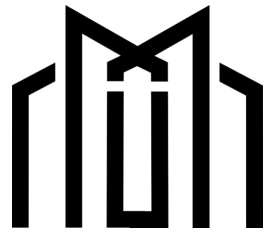


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UNIVERSAL ELECTRONICS BOX

Version 01.18.2026



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The components and instructions in this manual can cause serious injury, fire, or damage to equipment if handled improperly. Power supplies, motor drivers, wiring, and control boards may carry hazardous voltages and currents even when the machine appears to be powered off. Always disconnect mains power before installing, modifying, or servicing any electrical components. Use properly rated wires, connectors, fuses, and grounding methods, and never bypass safety features or protective devices. Incorrect wiring, loose connections, or inadequate insulation can lead to electric shock, overheating, or fire. **If there is anything, and we mean anything that you are curious or unsure about, you are more than welcome to ask us on our discord server.**

The Universal Electronics Box (UBE) is our effort to standardize on a set of higher-performance, more reliable, and safer electronic components across all of our machines. The wiring diagram shown on page 22 represents our standard electrical layout and should be followed for all builds, even if you are installing the electronics in a Casa or another enclosure. Standardizing the wiring helps reduce errors, improves safety, and makes troubleshooting and future upgrades easier.

INTRODUCTION

FRAME ASSEMBLY

FRAME.....

COVERS

ELECTRONICS PRE-ASSEMBLY

ELECTRONICS.....

WIRING

ELECTRONICS.....

ASSEMBLY

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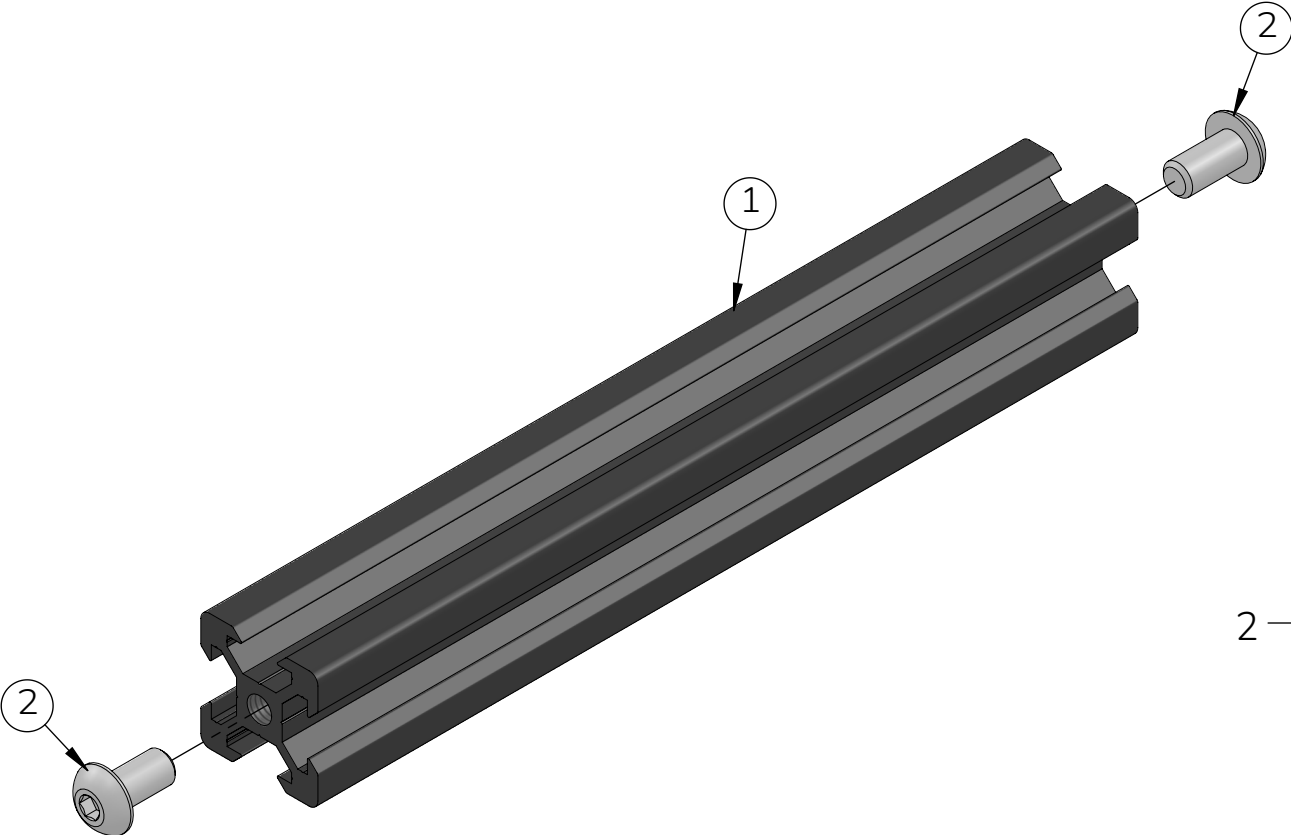
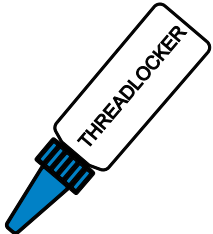
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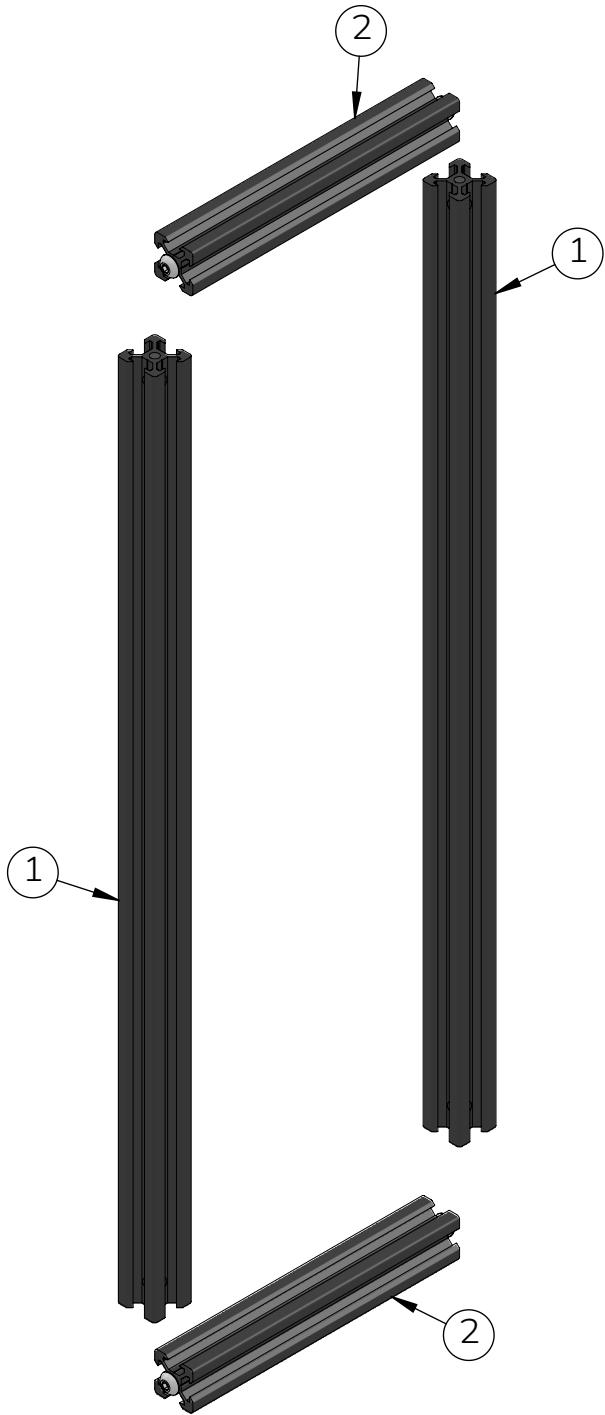
34

QTY 2

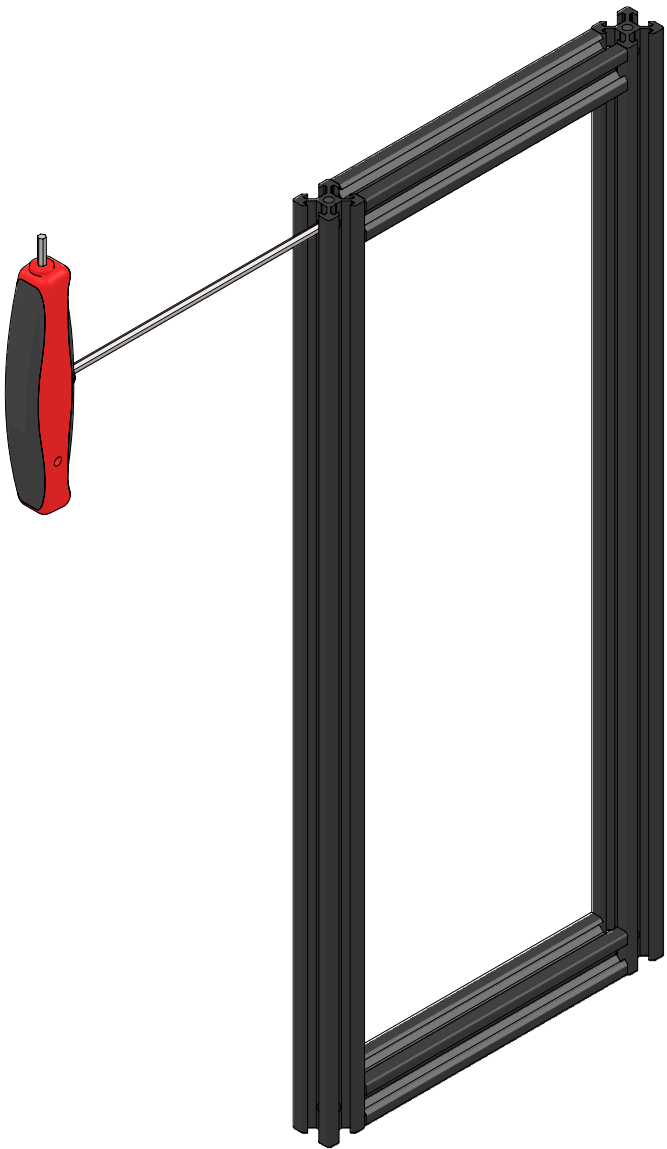
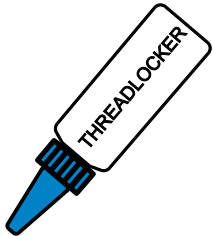
	PART NUMBER	QTY.
1	2020-V-SLOT-140mm	1
2	ISO-7380-M5X10 BHCS	2



QTY 2



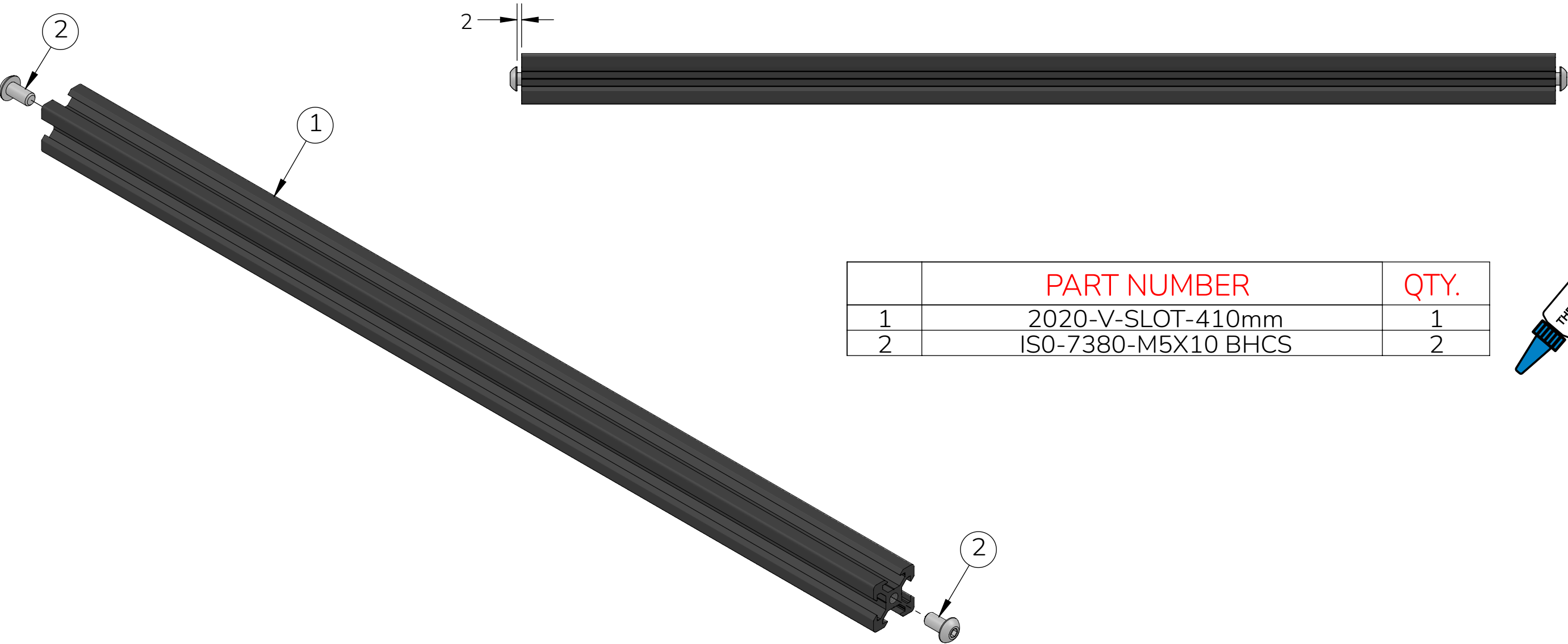
	PART NUMBER	QTY.
1	2020-V-SLOT-430mm	2
2	2020-V-SLOT-140mm-ASSEMBLY	2



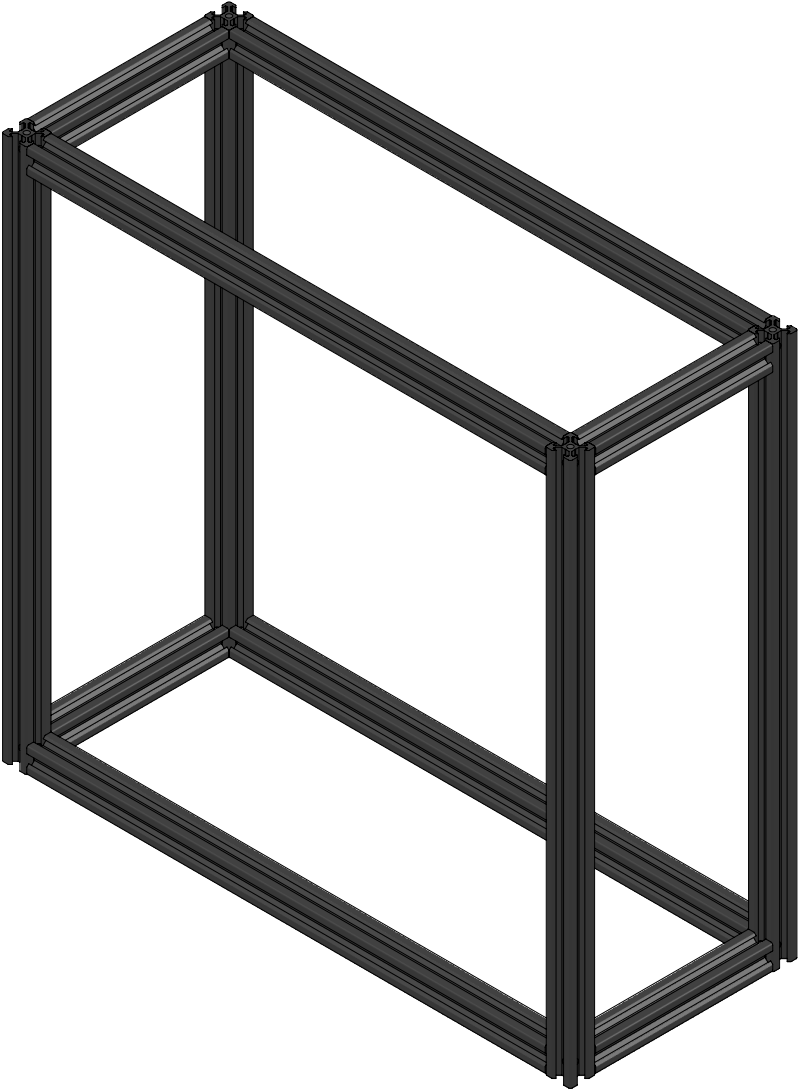
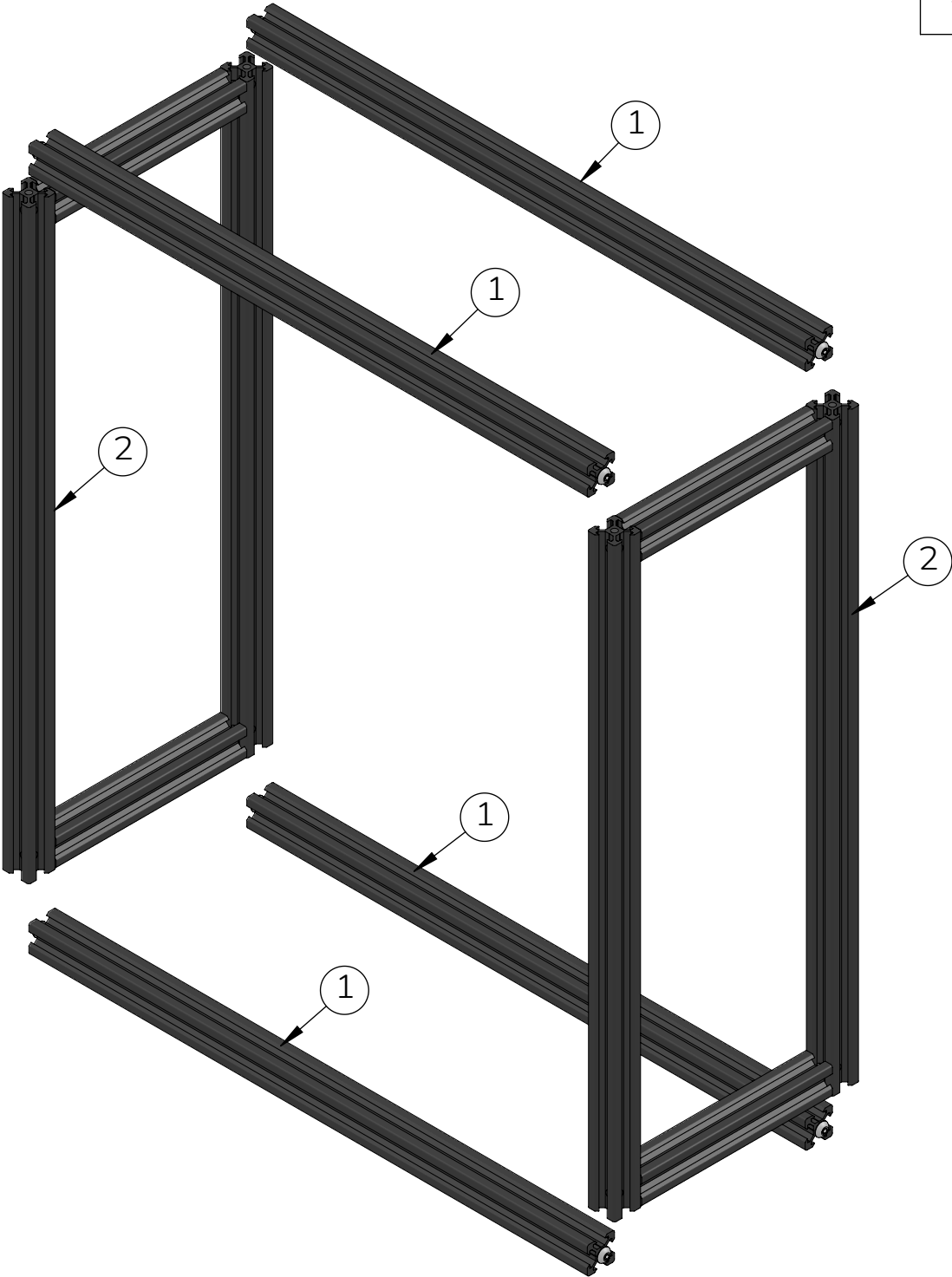
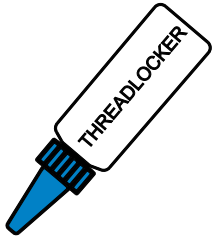
FRAME ASSEMBLY

FRONT REAR V-SLOT ASSEMBLY

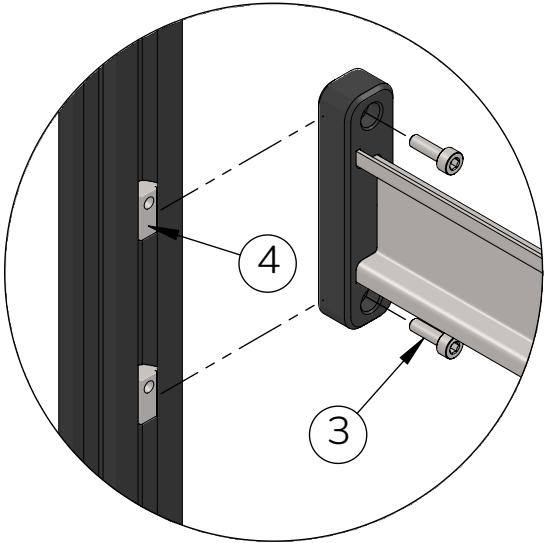
QTY 2



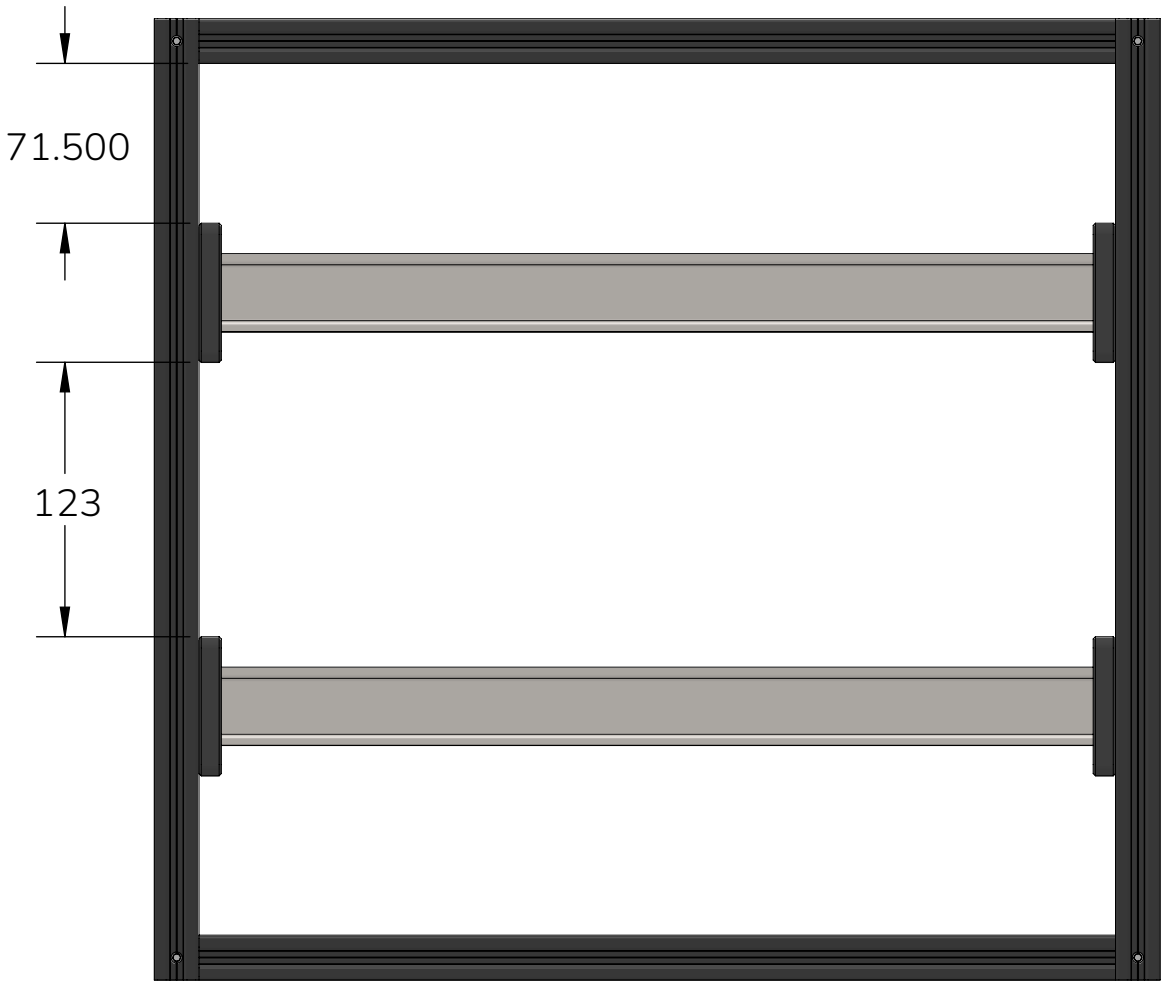
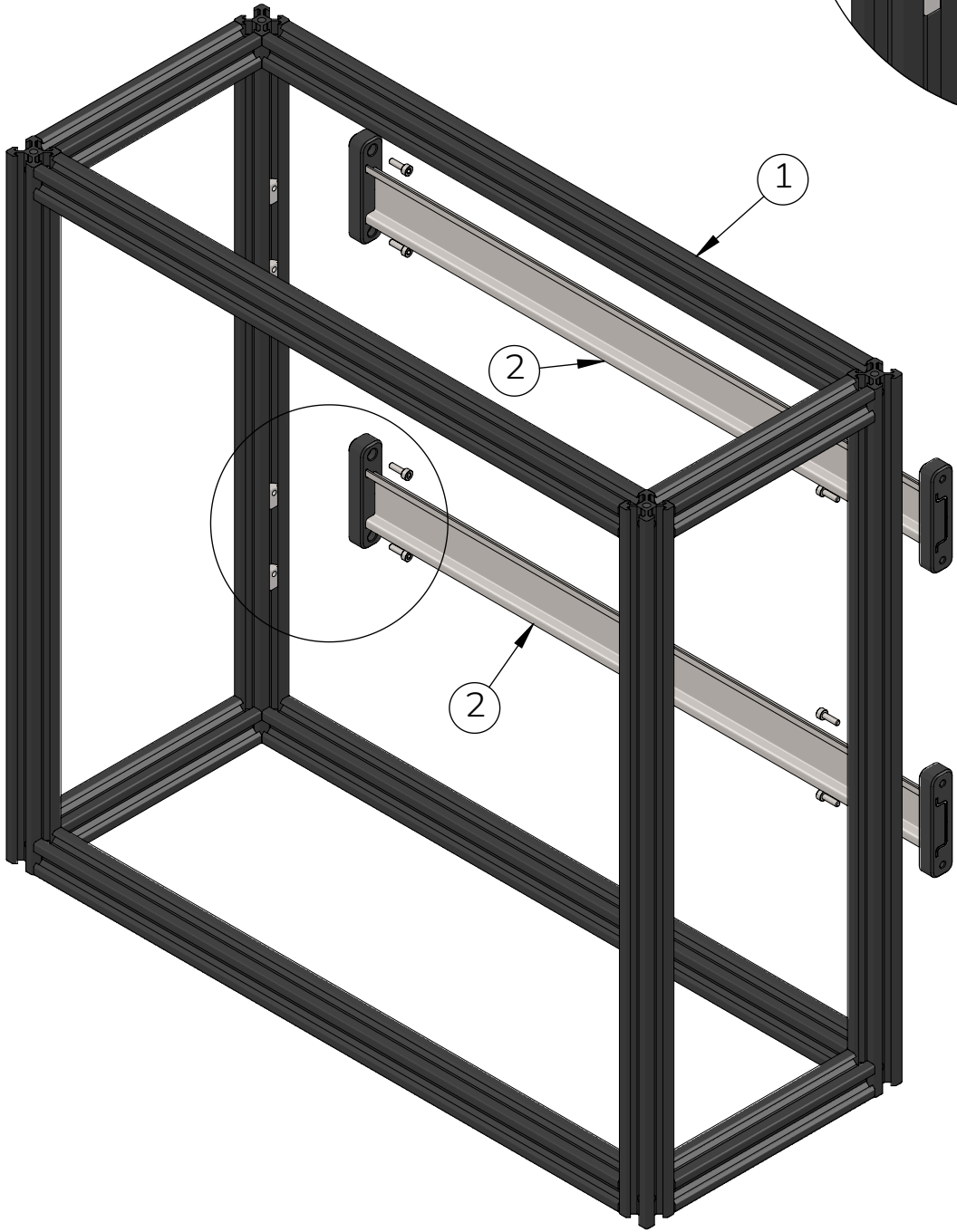
	PART NUMBER	QTY.
1	2020-V-SLOT-410mm-ASSEMBLY	4
2	FRAME-FRONT-REAR-V-SLOT-ASSEMBLY	2



FRAME ASSEMBLY

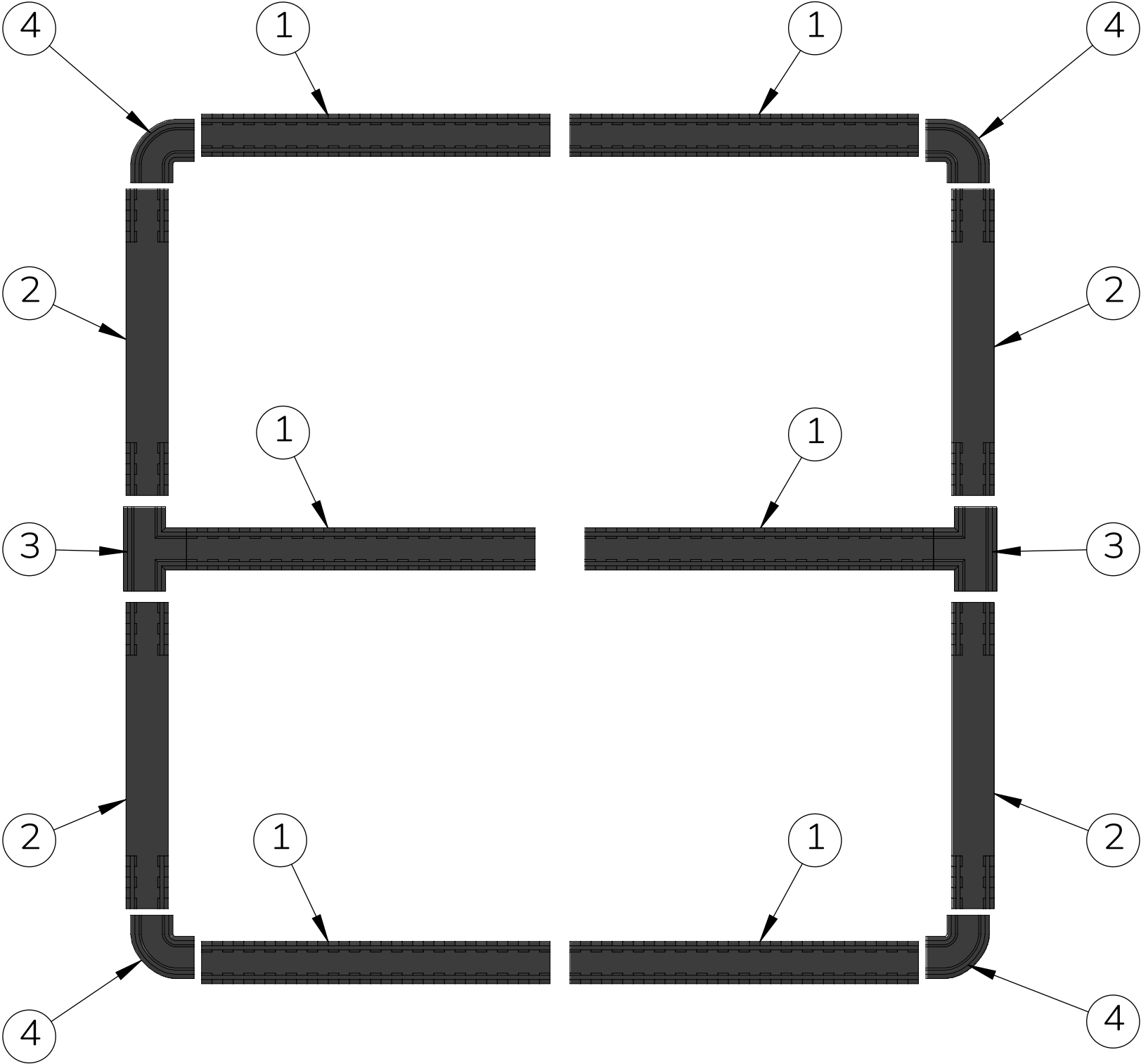


	PART NUMBER	QTY.
1	EBOX-FRAME-ASSEMBLY	1
2	DIN-RAIL-MOUNT-ASSEMBLY	2
3	ISO-4762-M3X10-SHCS	8
4	M3 DROP IN NUT	8



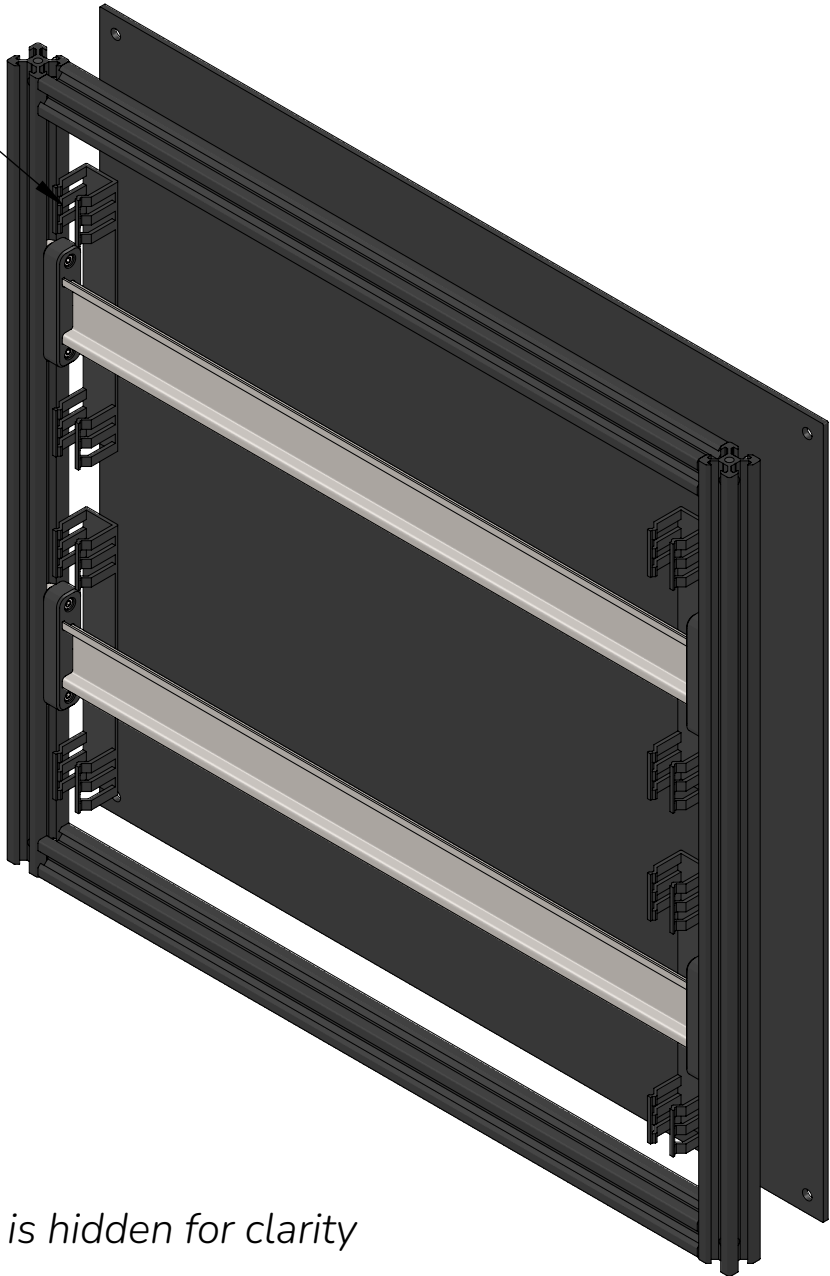
	PART NUMBER	QTY.
1	CABLE-DUCT-165mm	6
2	CABLE-DUCT-145mm	4
3	CABLE-DUCT-T-JUNCTION	2
4	CABLE-DUCT-ELBOW	4

Don't start putting this together yet, we are going to assemble this in the EBox Frame so that we know our alignment is perfect.

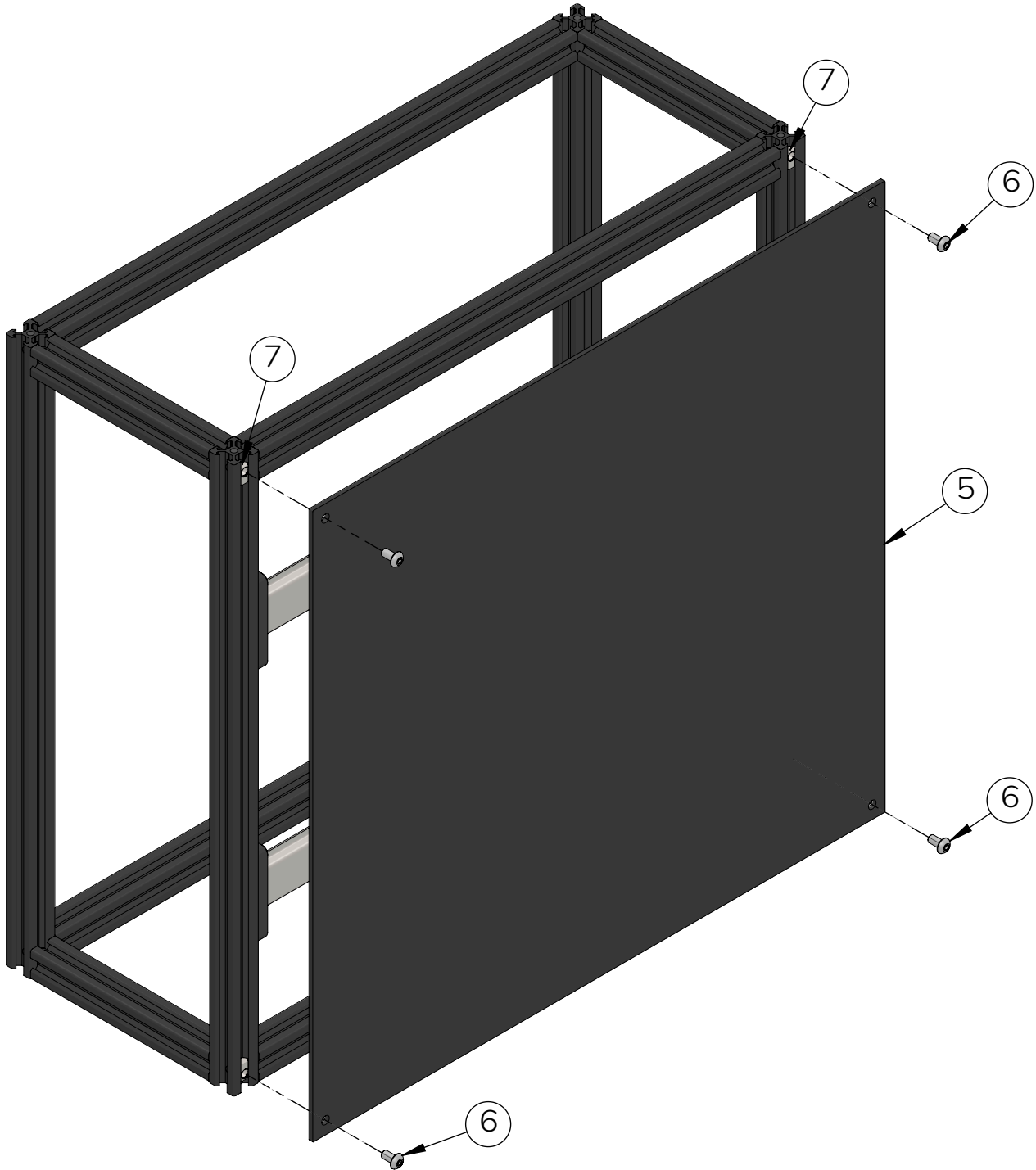


	PART NUMBER	QTY.
1	EBOX-FRAME-ASSEMBLY	1
5	450X430X3MM-PANEL	1
6	ISO-7380-M5X10 BHCS	4
7	M5 DROP IN NUT	4

PLACE THE (4) 145mm CABLE
DUCT PIECES BETWEEN THE
DIN RAIL AND THE PANEL

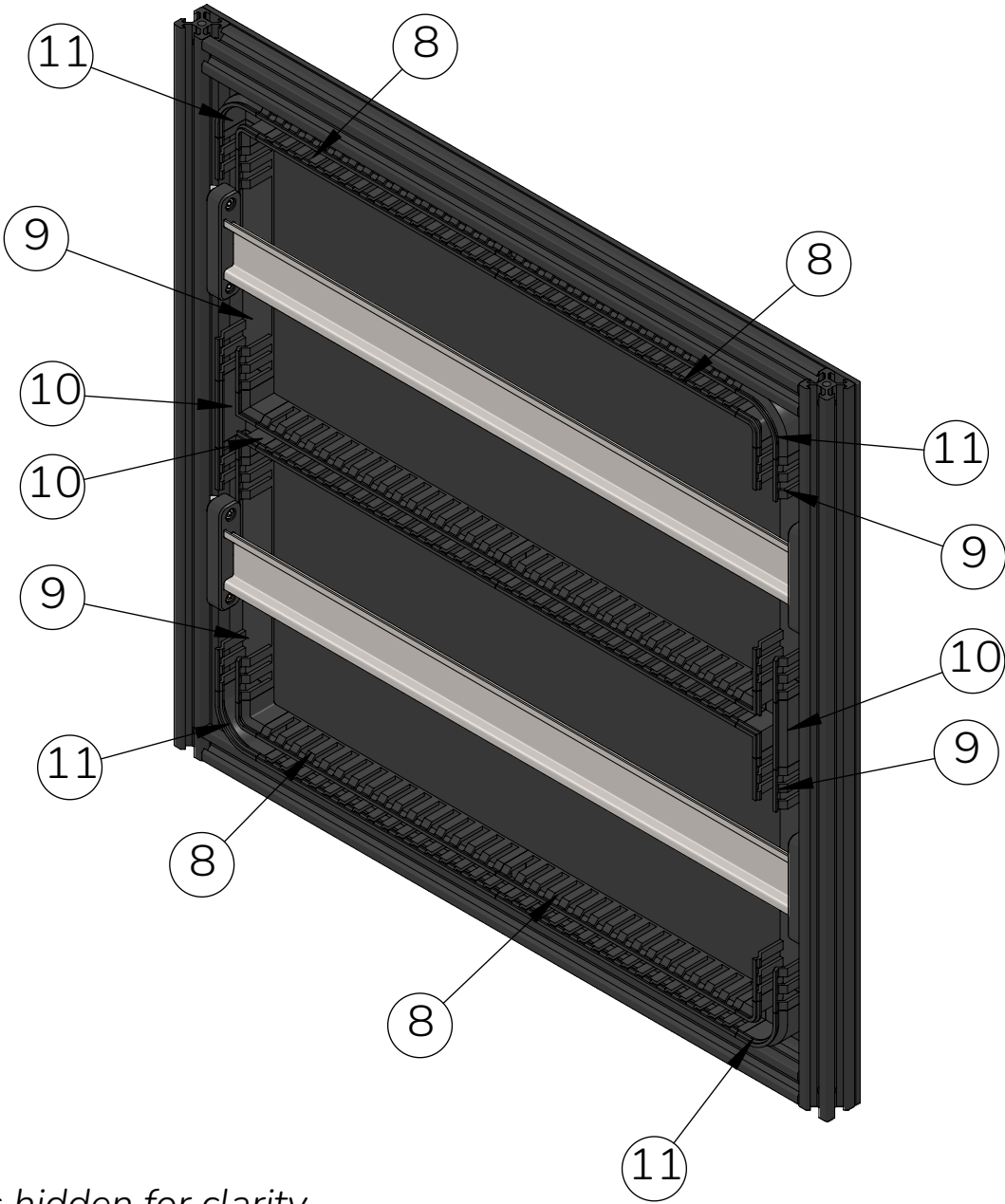


Part of the Frame is hidden for clarity



	PART NUMBER	QTY.
8	CABLE-DUCT-165mm	6
9	CABLE-DUCT-145mm	4
10	CABLE-DUCT-T-JUNCTION	2
11	CABLE-DUCT-ELBOW	4

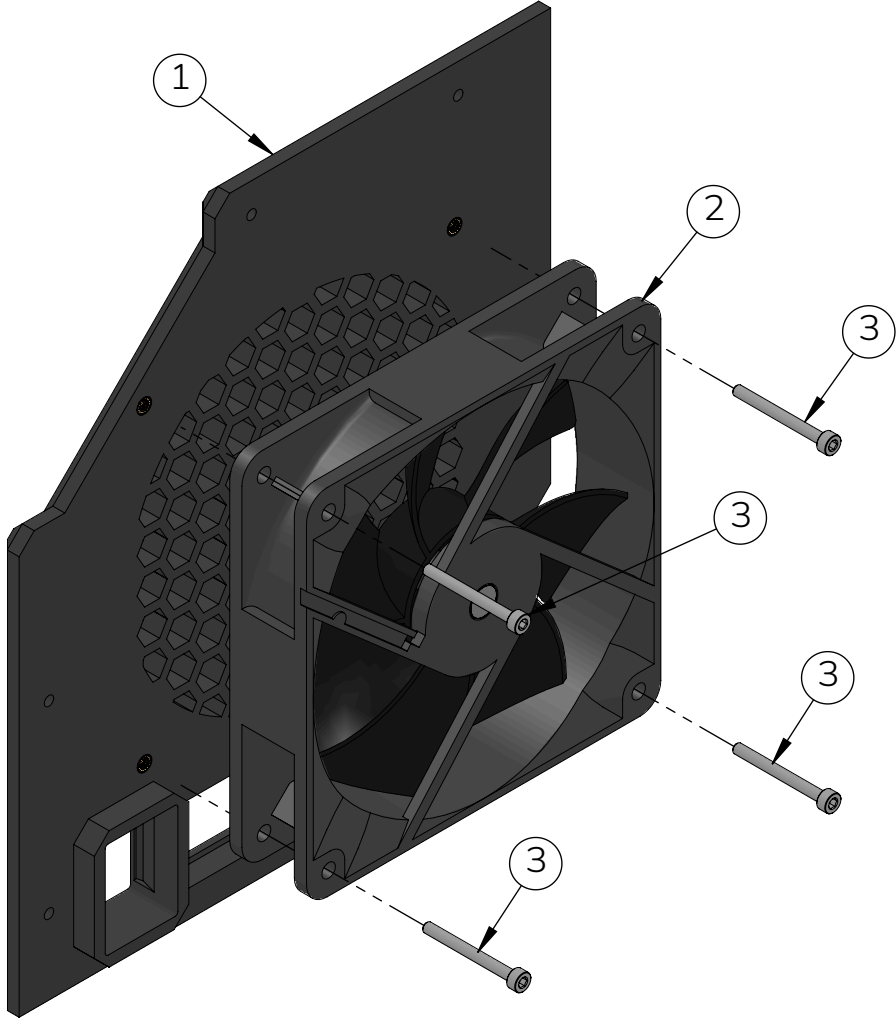
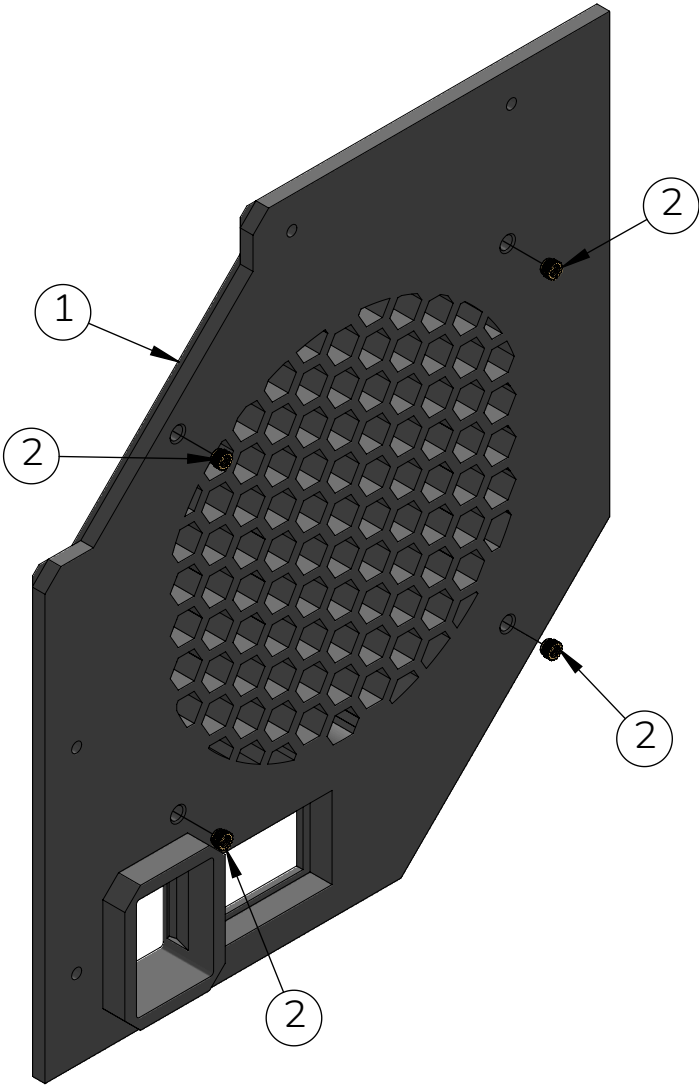
VHB Tape the back of each piece after fitting and aligning.



Part of the Frame is hidden for clarity

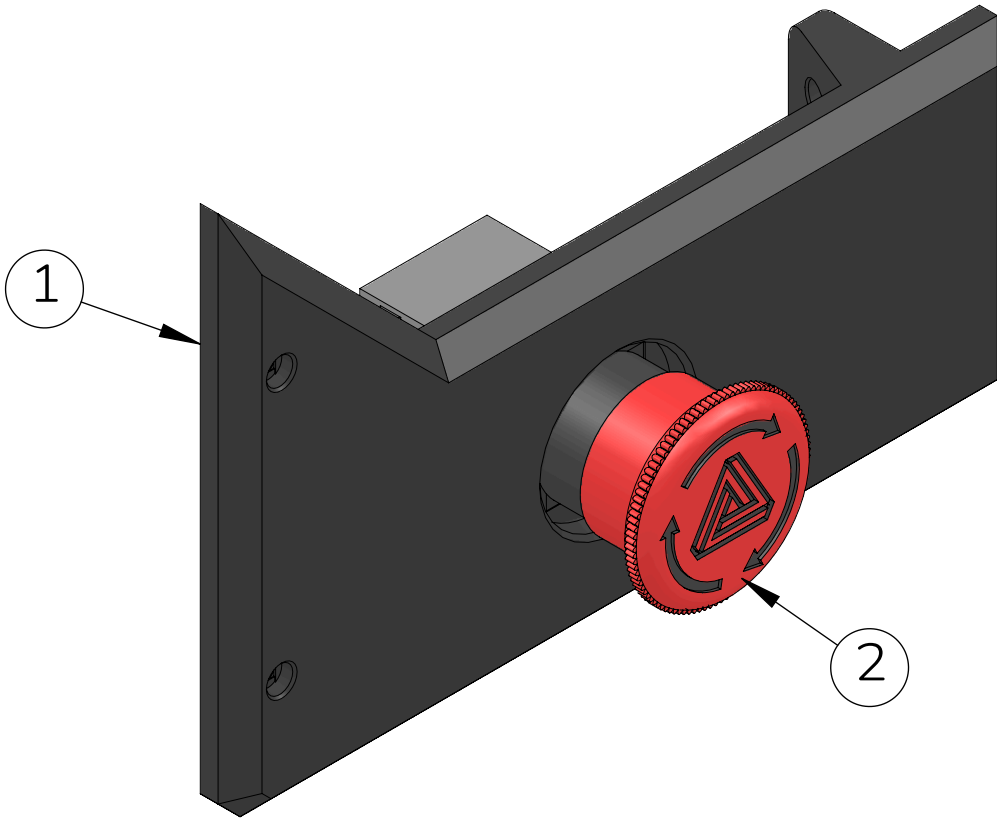
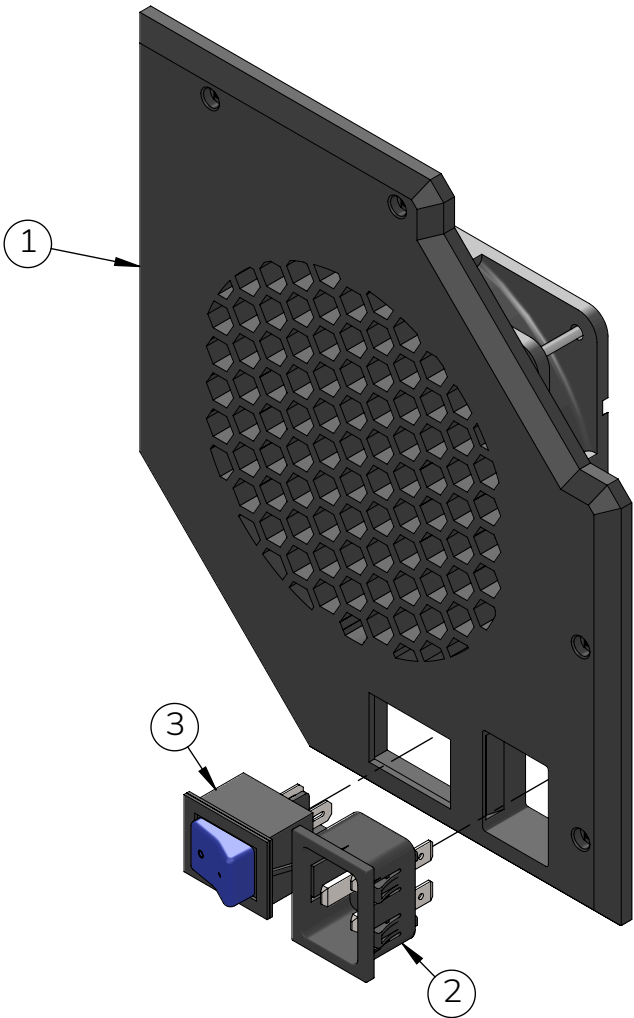
	PART NUMBER	QTY.
1	REAR-INLET-COVER	1
2	M3X3MM HEATSET INSERT	4

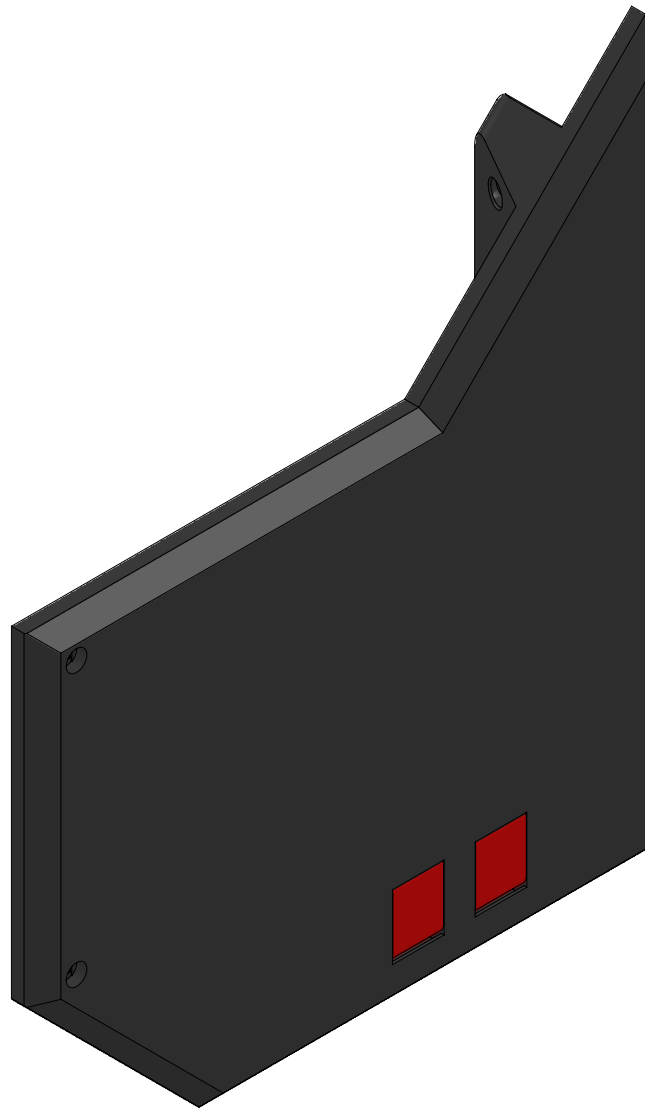
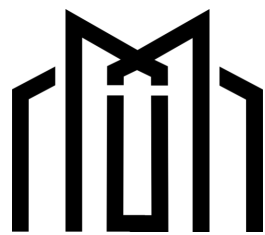
	PART NUMBER	QTY.
1	REAR-INLET-COVER PRE-ASSEMBLY	1
2	120mm FAN	1
3	ISO-4762-M3X30-SHCS	4



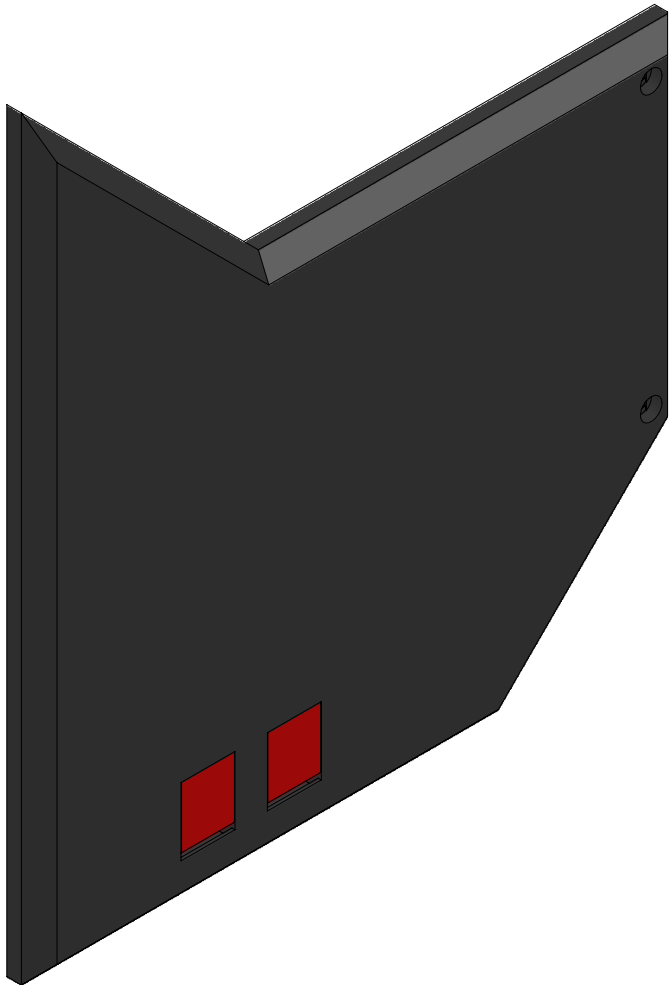
	PART NUMBER	QTY.
1	REAR-INLET-COVER PRE-ASSEMBLY	1
2	C20-INLET	1
3	KCD4-ROCKER-SWITCH	1

	PART NUMBER	QTY.
1	FRONT-ESTOP-COVER-34mm	1
2	E-STOP	1



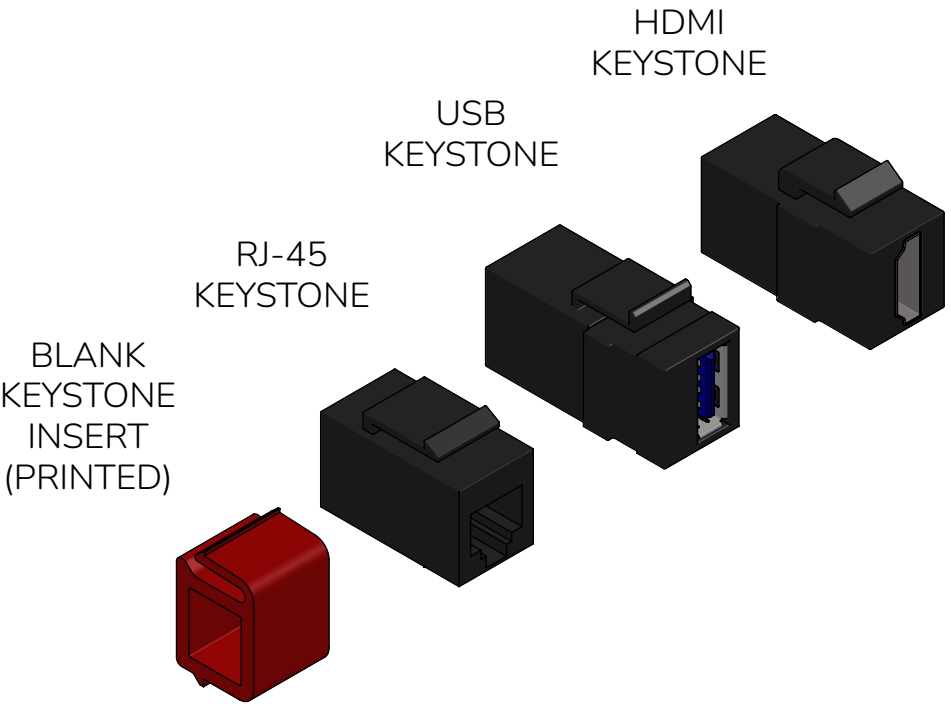


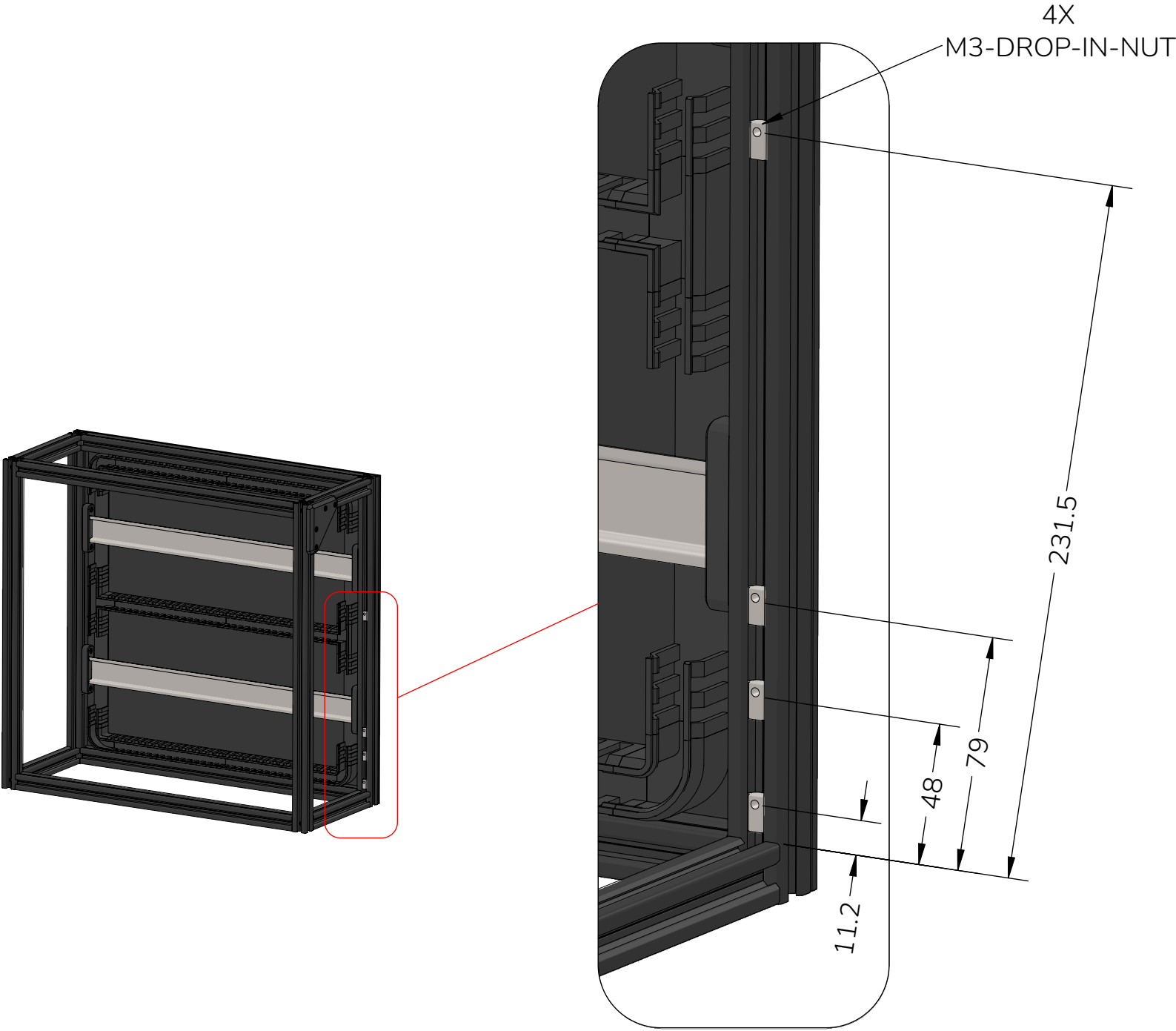
FRONT MID COVER



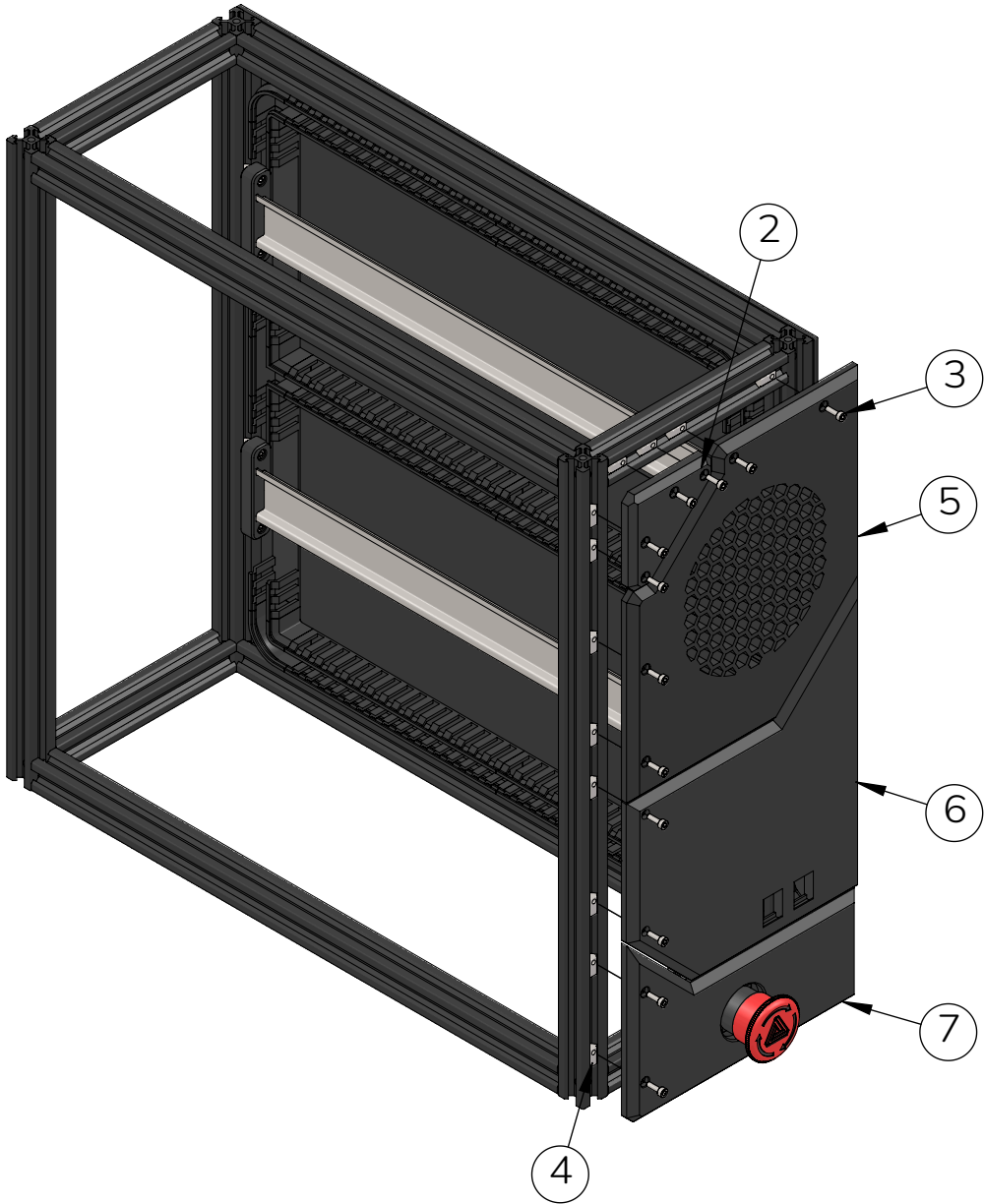
FRONT MID COVER

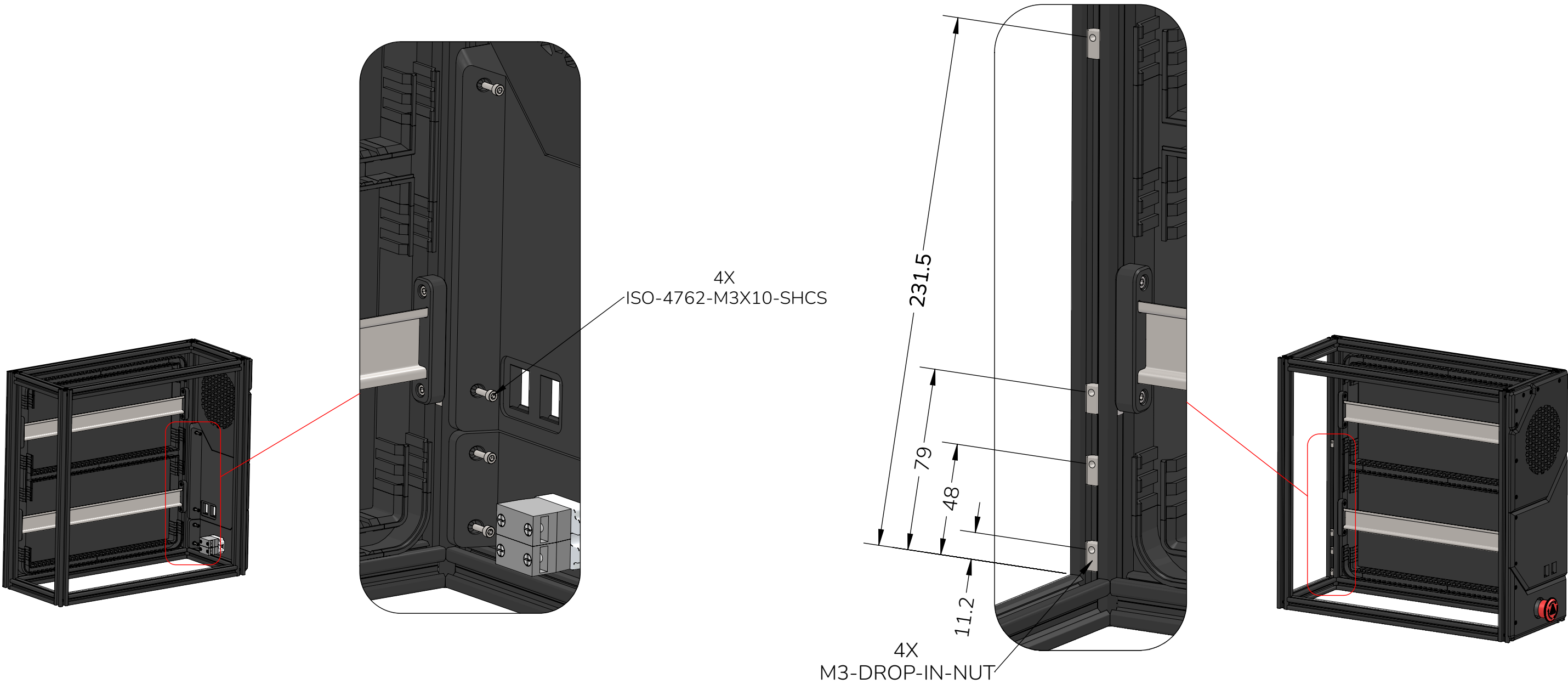
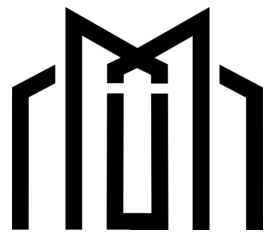
KEYSTONES CAN BE PURCHASED TO PASS THROUGH USB, ETHERNET (RJ45), OR HDMI. PRINTED BLANK INSERTS ARE PROVIDED

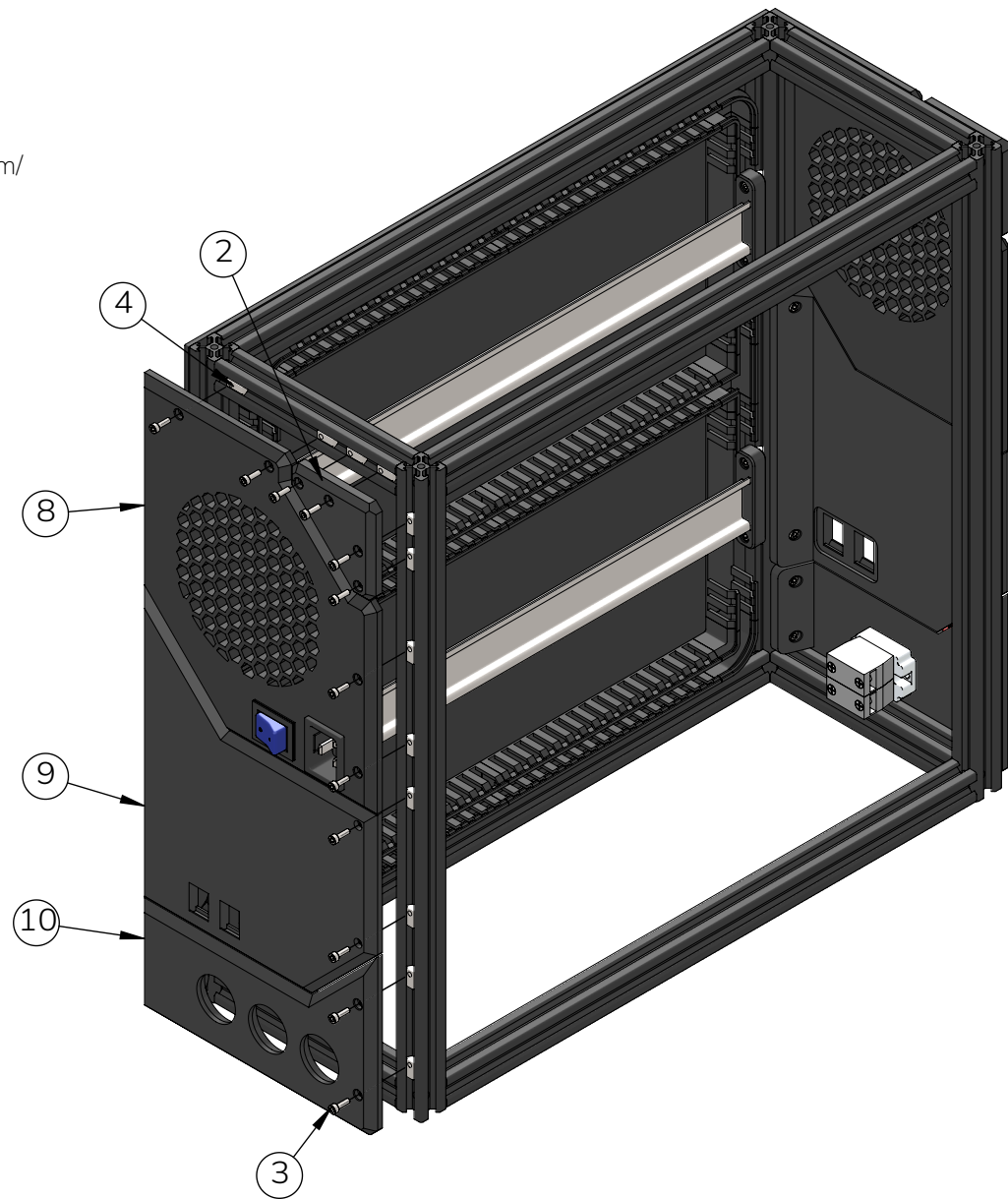
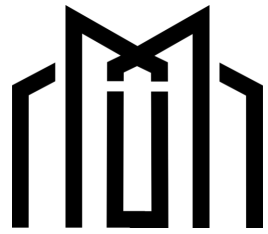




	PART NUMBER	QTY.
2	PANEL-CLIP-F	1
3	ISO-4762-M3X10-SHCS	12
4	M3 DROP IN NUT	12
5	FRONT-INLET-COVER	1
6	FRONT-MID-COVER	1
7	FRONT-ESTOP-COVER-ASSEMBLY	1

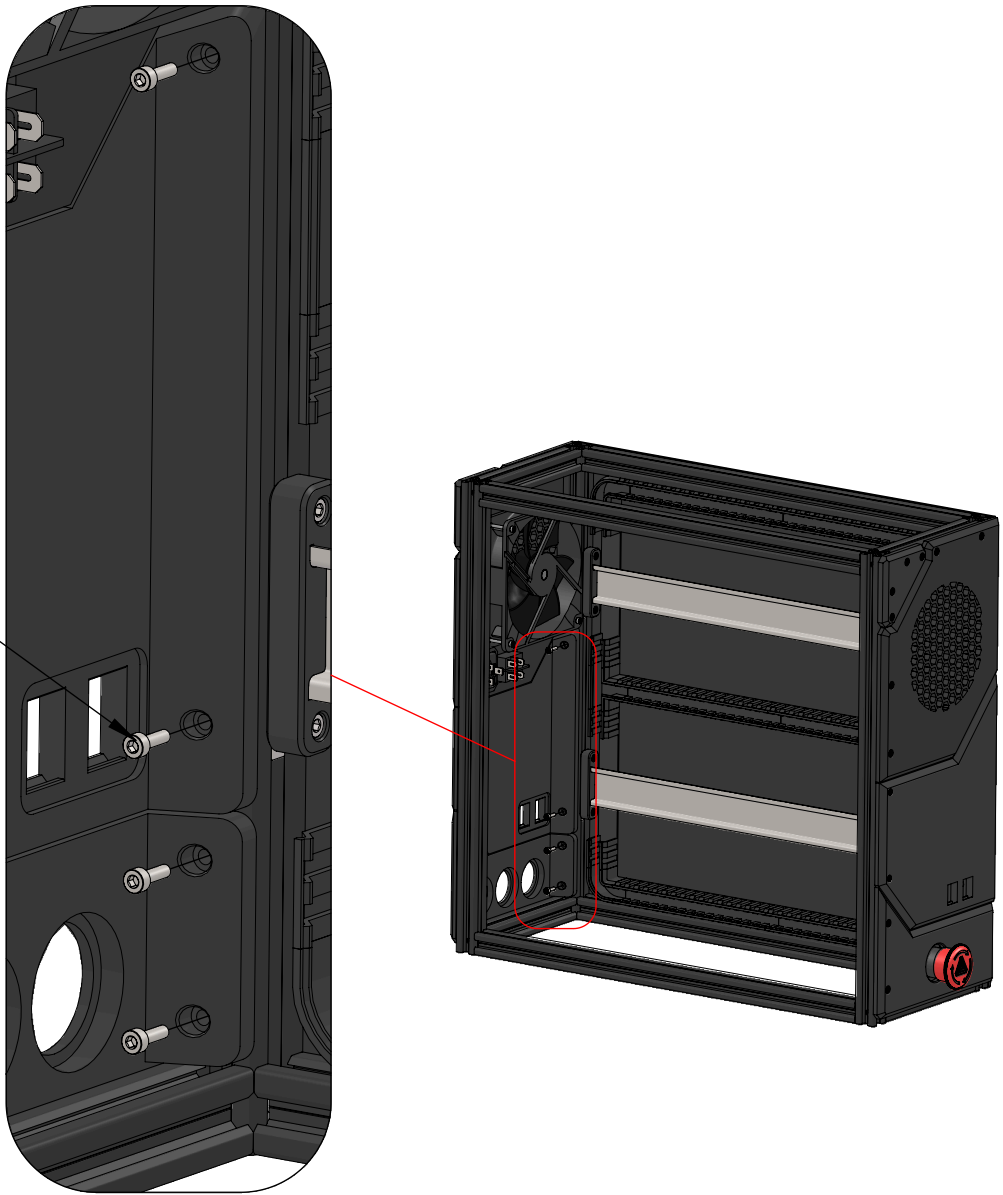




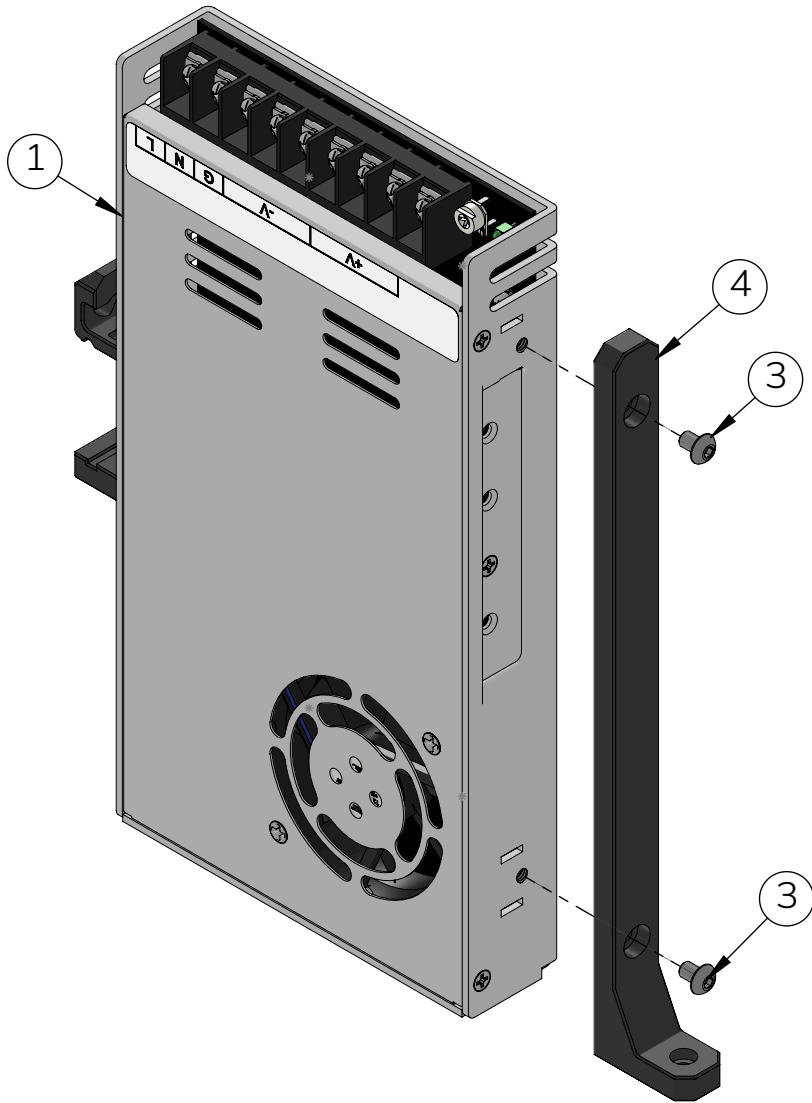
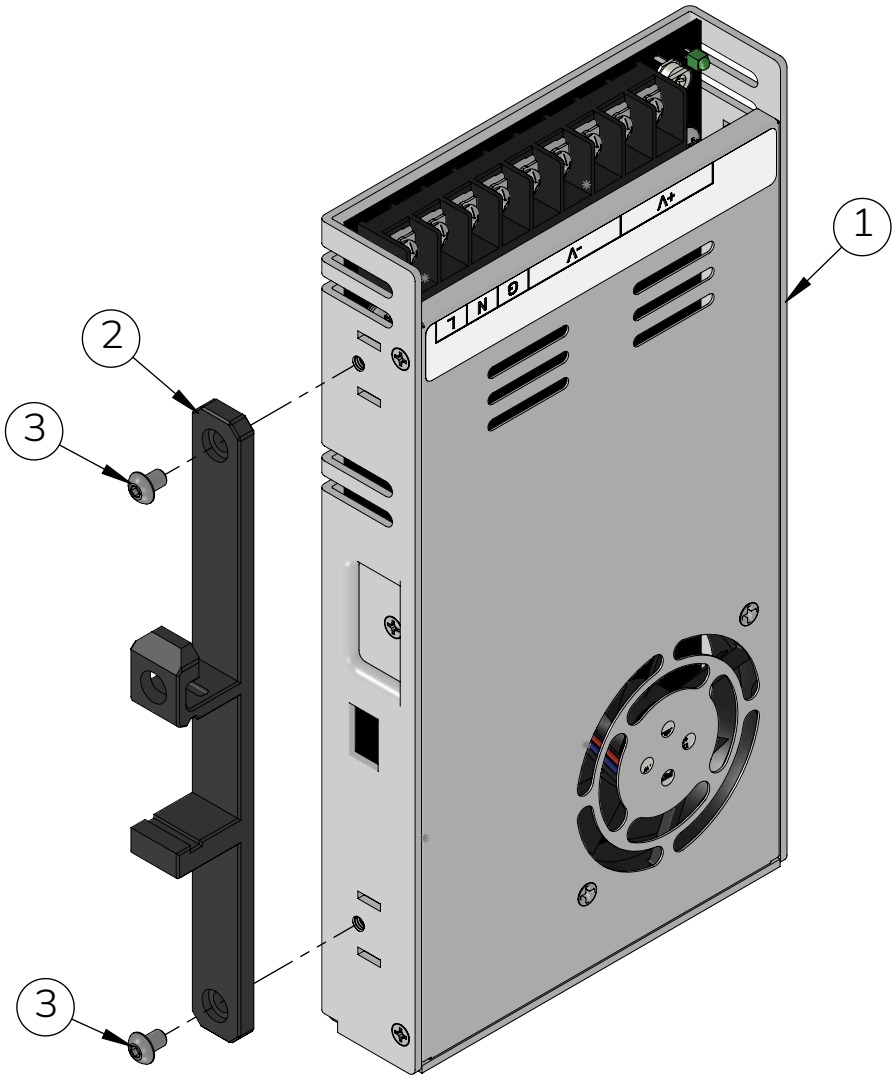


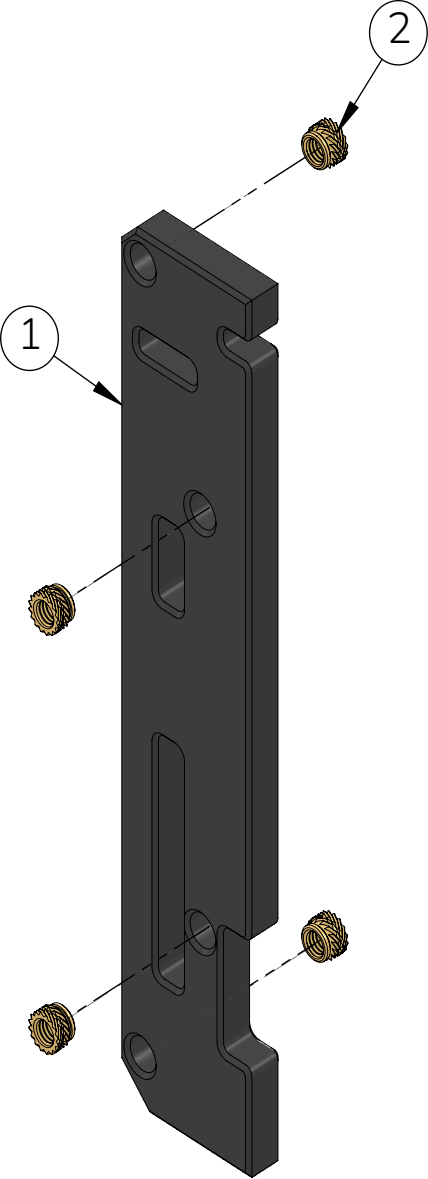
	PART NUMBER	QTY.
2	PANEL-CLIP-F	1
3	ISO-4762-M3X10-SHCS	12
4	M3 DROP IN NUT	12
8	REAR-INLET-COVER-ASSEMBLY	1
9	REAR-MID-COVER	1
10	REAR-LOWER-COVER	1

4X
ISO-4762-M3X10-SHCS

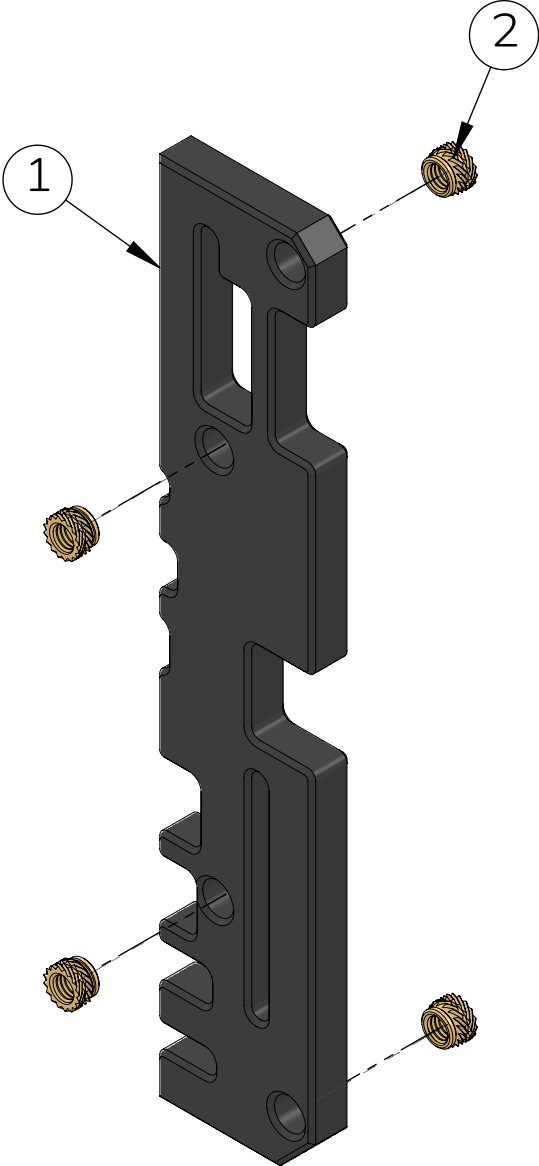


	PART NUMBER	QTY.
1	LRS-350	1
2	LRS-350-DIN-MOUNT	1
3	ISO-7380-M4X6-BHCS	4
4	LRS-350-GUSSET	1



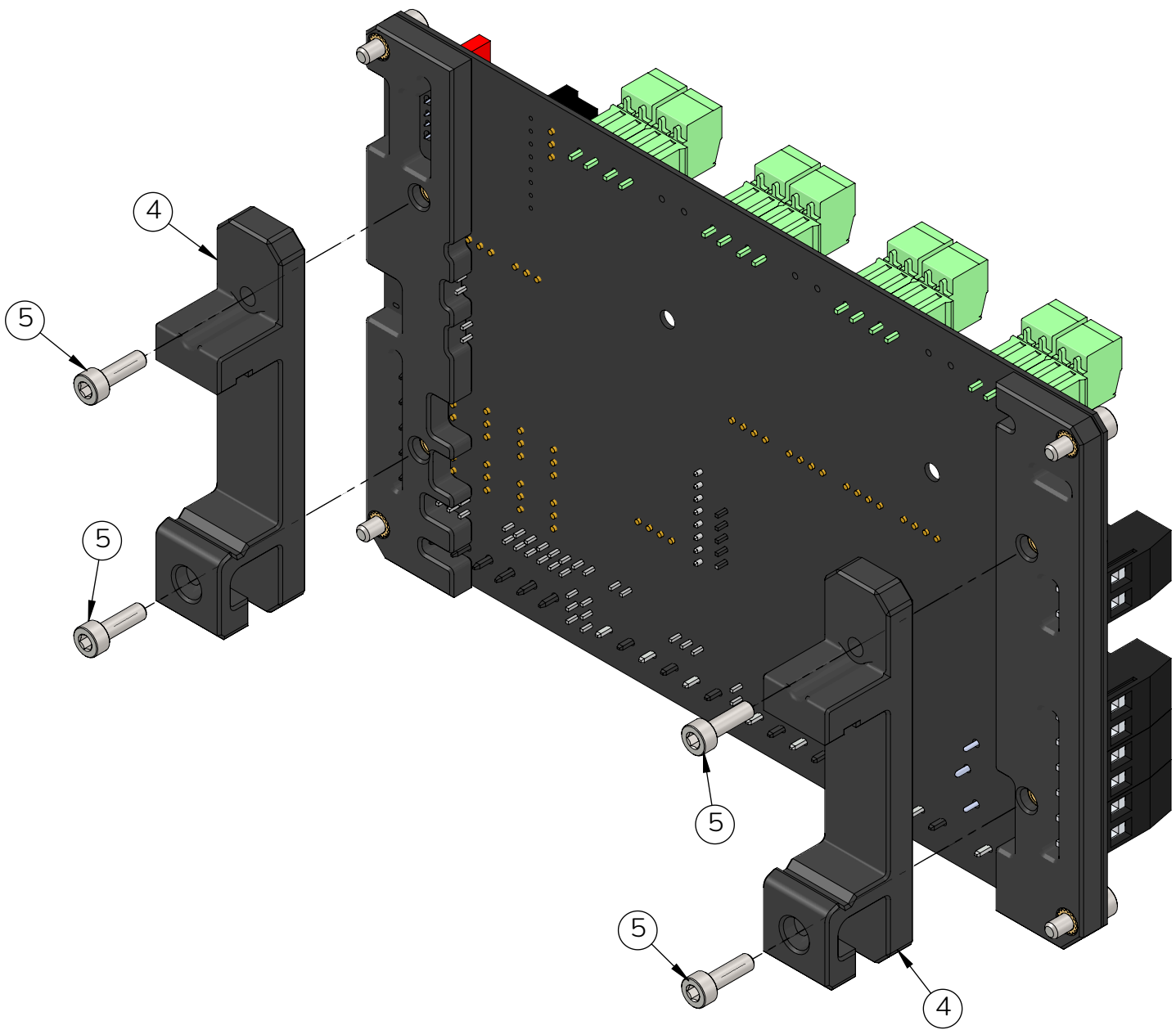
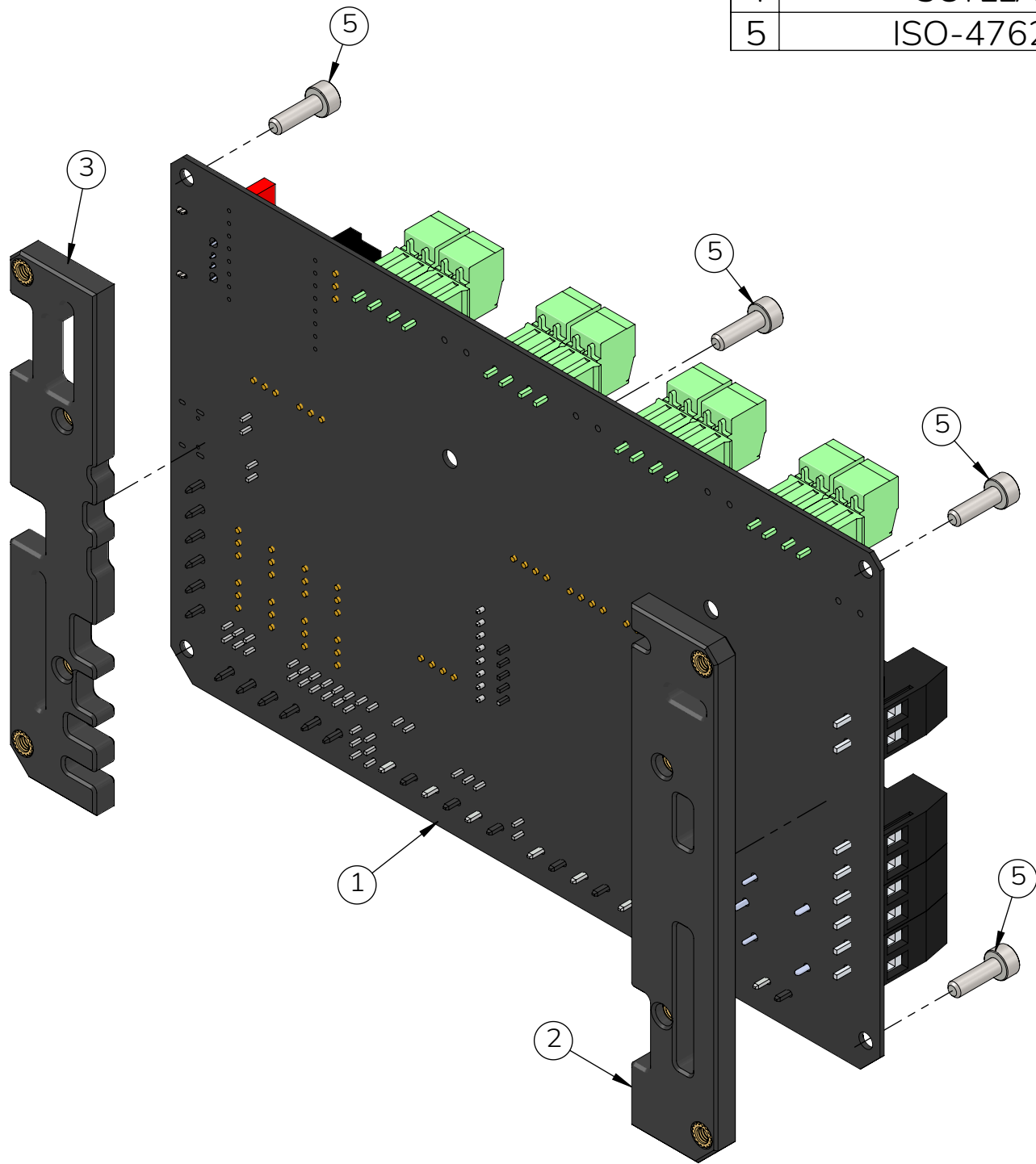


	PART NUMBER	QTY.
1	SCYLLA-DIN-ADAPTER-RIGHT	1
2	M3X3MM HEATSET INSERT	4

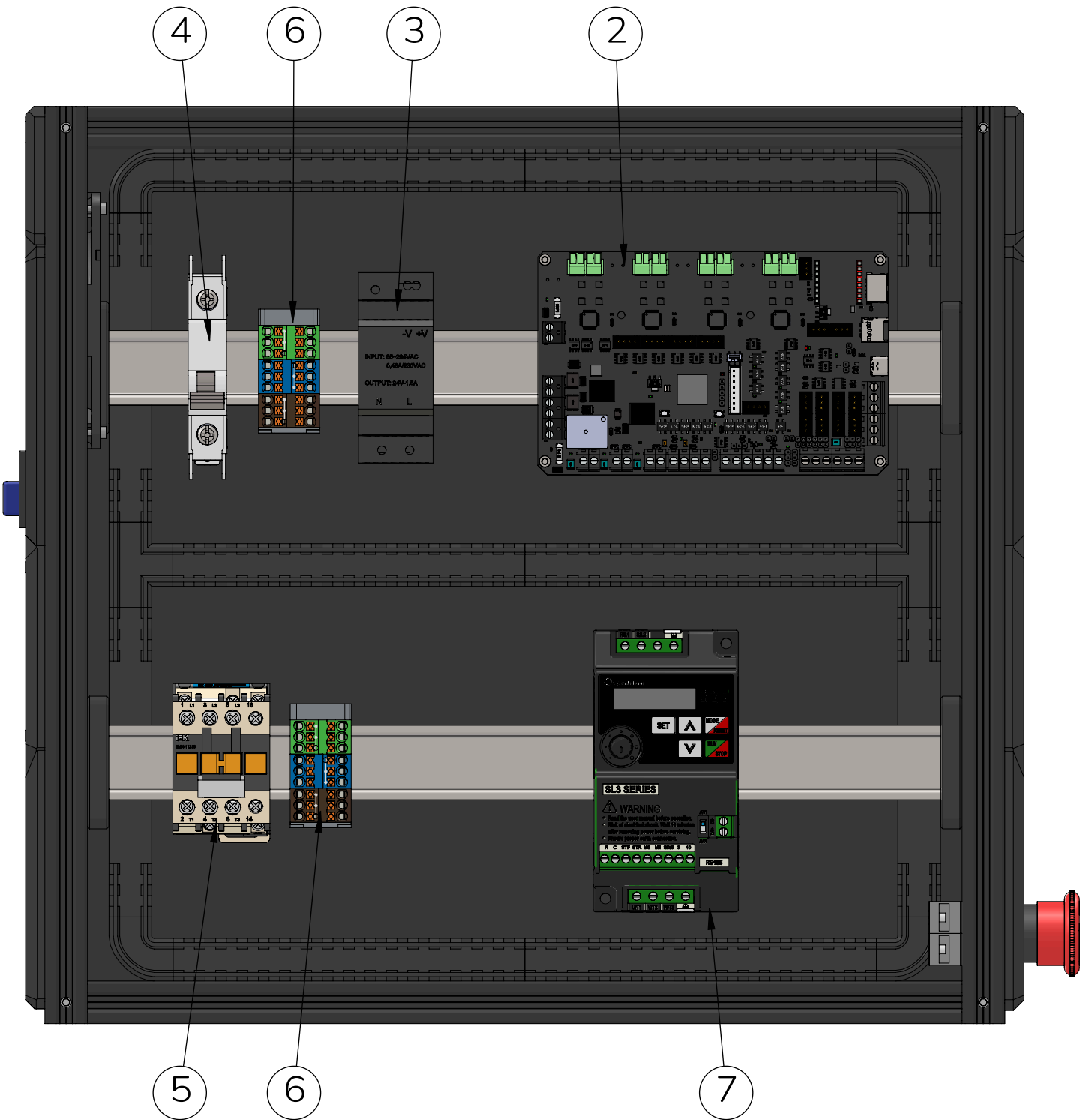


	PART NUMBER	QTY.
1	SCYLLA-DIN-ADAPTER-LEFT	1
2	M3X3MM HEATSET INSERT	4

	PART NUMBER	QTY.
1	BIGTREE TECH SCYLLA V1.0	1
2	SCYLLA-MOUNT-A-PRE-ASSEMBLY	1
3	SCYLLA-MOUNT-B-PRE-ASSEMBLY	1
4	SCYLLA-DIN-MOUNT	2
5	ISO-4762-M3X10-SHCS	8



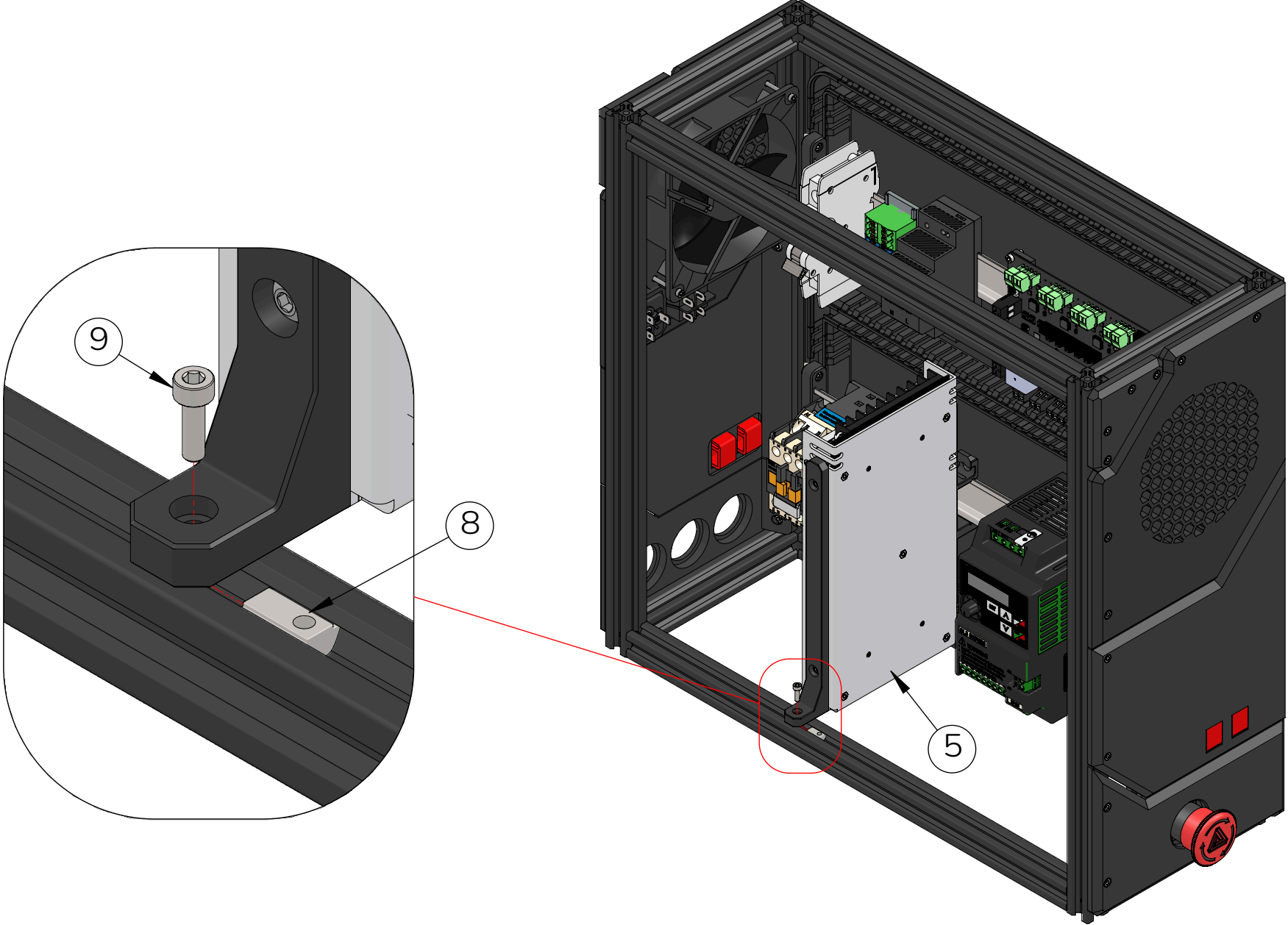
ELECTRONICS PRE-ASSEMBLY
SCYLLA PRE-ASSEMBLY

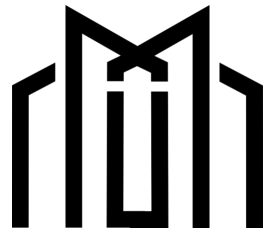


	PART NUMBER	QTY.
2	SCYLLA-ASSEMBLY	1
3	MEAN WELL HDR-30-24	1
4	CIRCUIT BREAKER	1
5	CONTACTOR	1
6	TERMINAL BLOCK SET	2
7	SL3 VFD	1

MOUNT THE ELECTRICAL COMPONENTS
GENERALLY IN THESE LOCATIONS. THE
FINAL POSITIONING CAN BE ADJUSTED
DURING WIRING.

	PART NUMBER	QTY.
5	LRS-ASSEMBLY	1
8	M3 DROP IN NUT	1
9	ISO-4762-M3X10-SHCS	1

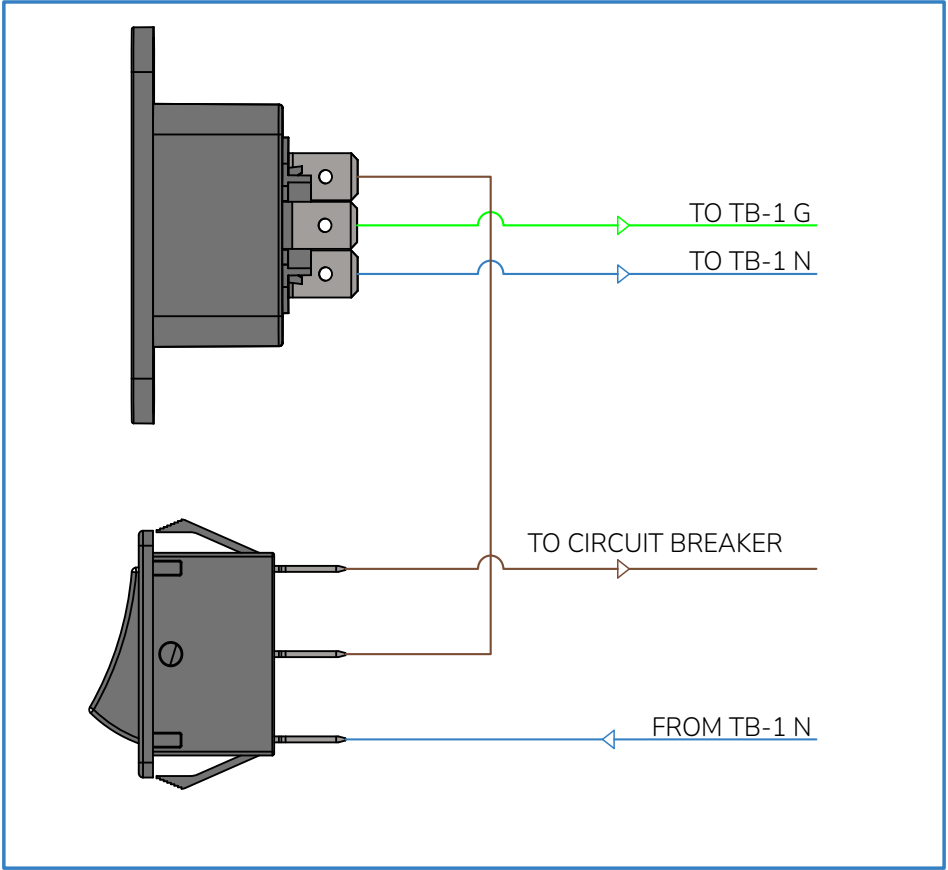




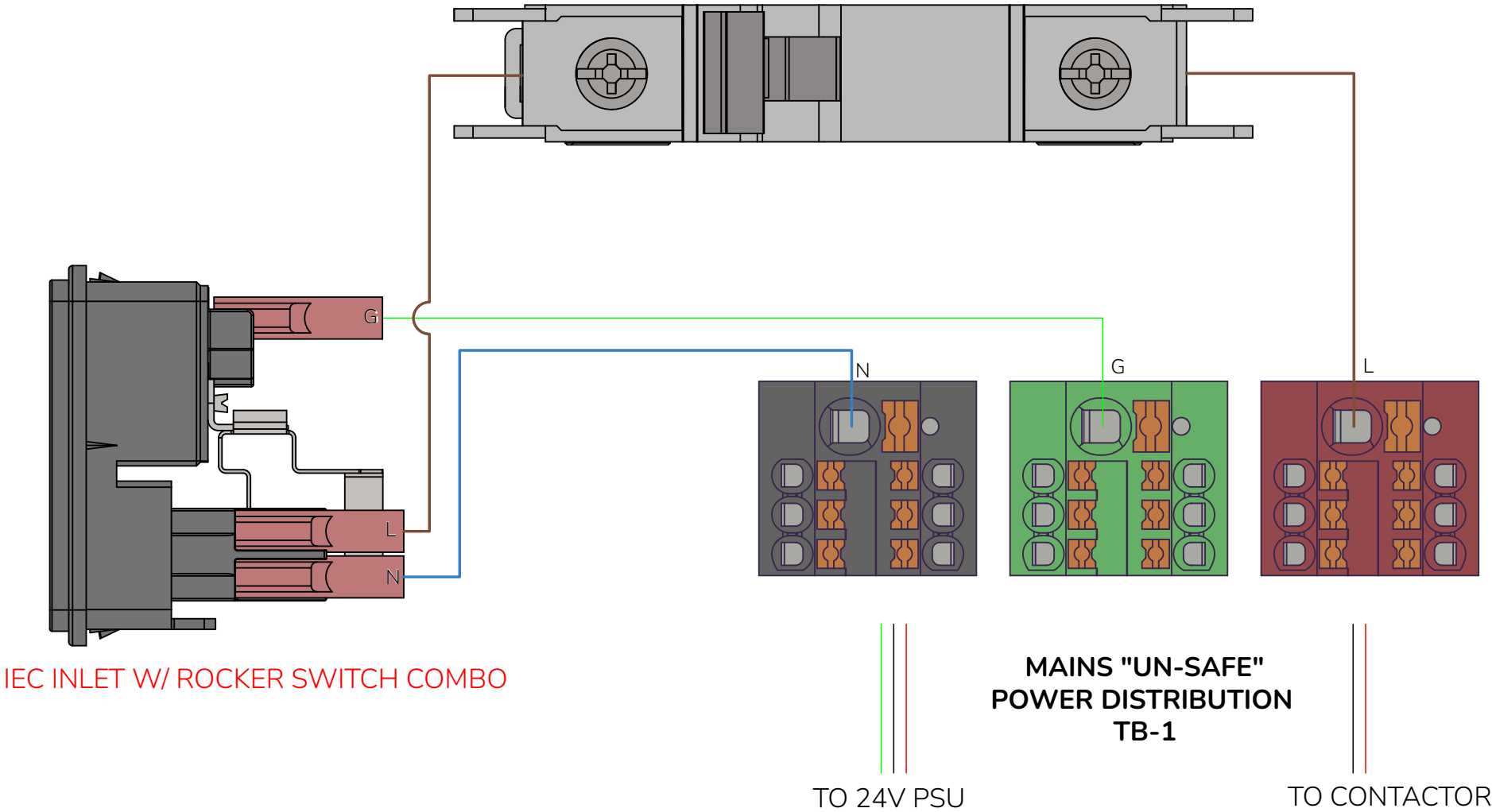
MAINS INLET WIRING

IEC color scheme is shown for the mains wiring. Depending on your region, the color scheme and wiring standards will differ.

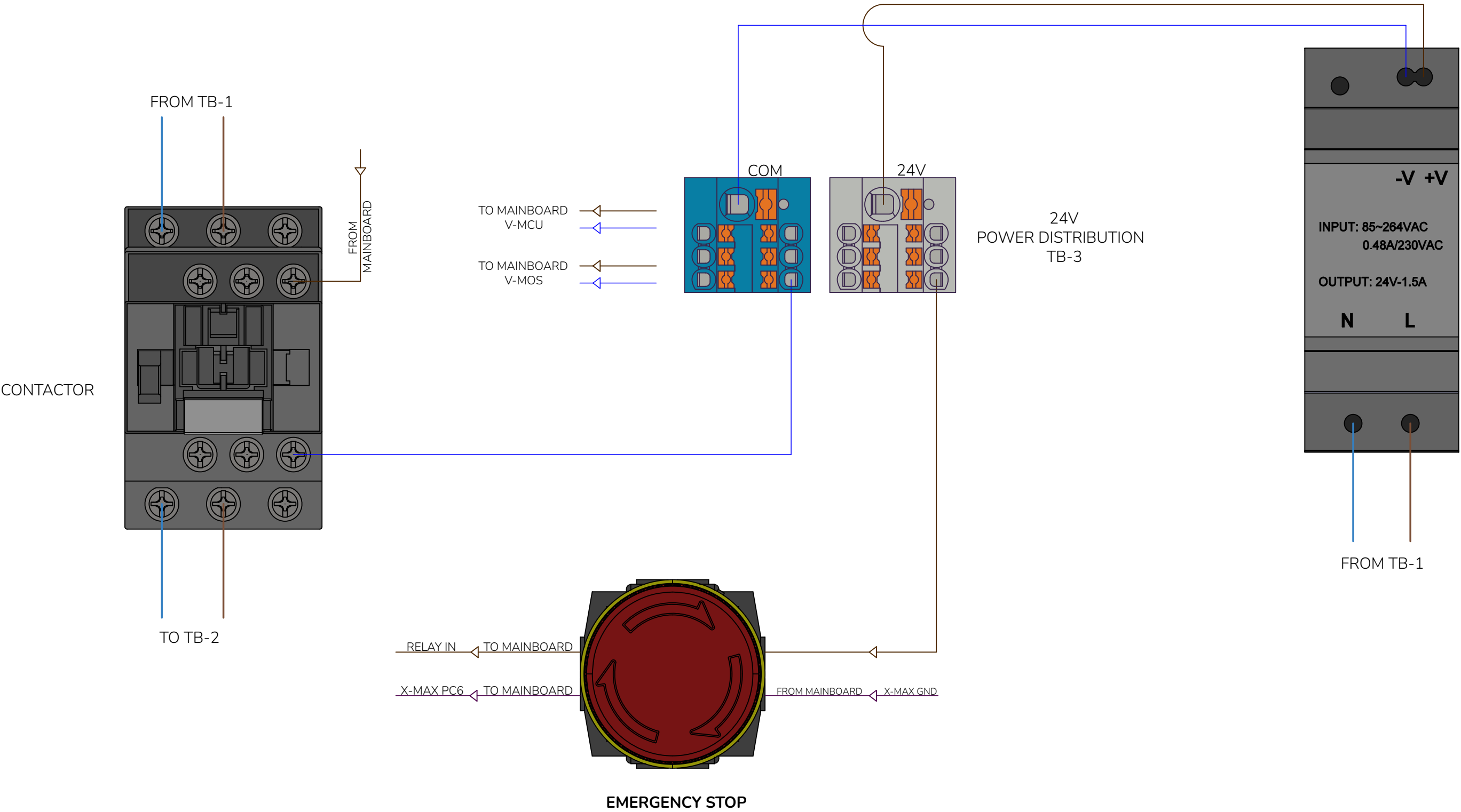
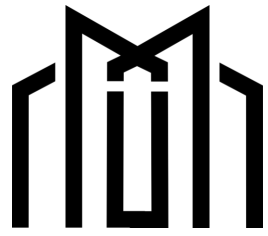
WIRING WITH INLET & ROCKER SWITCH



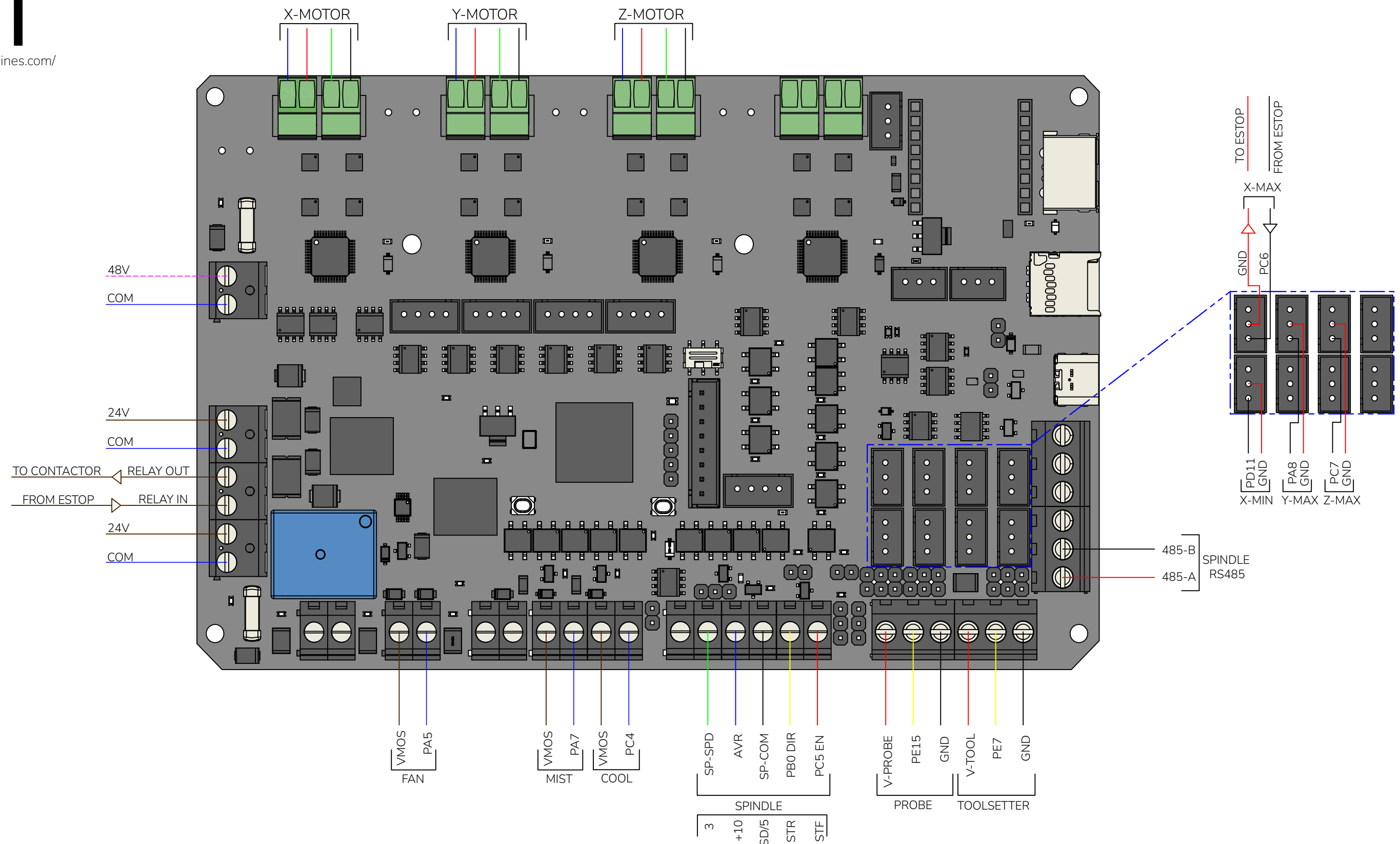
MINIATURE CIRCUIT BREAKER

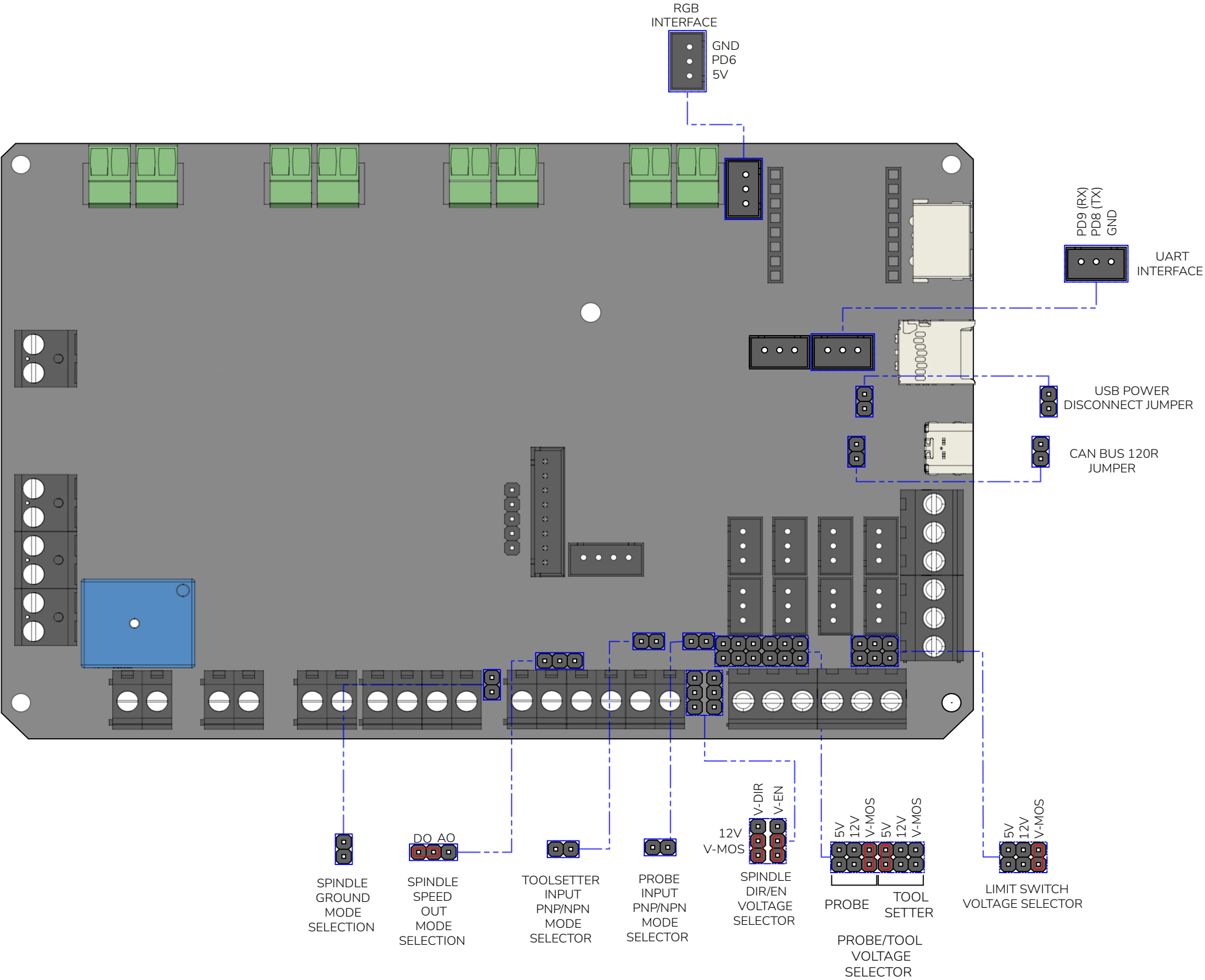


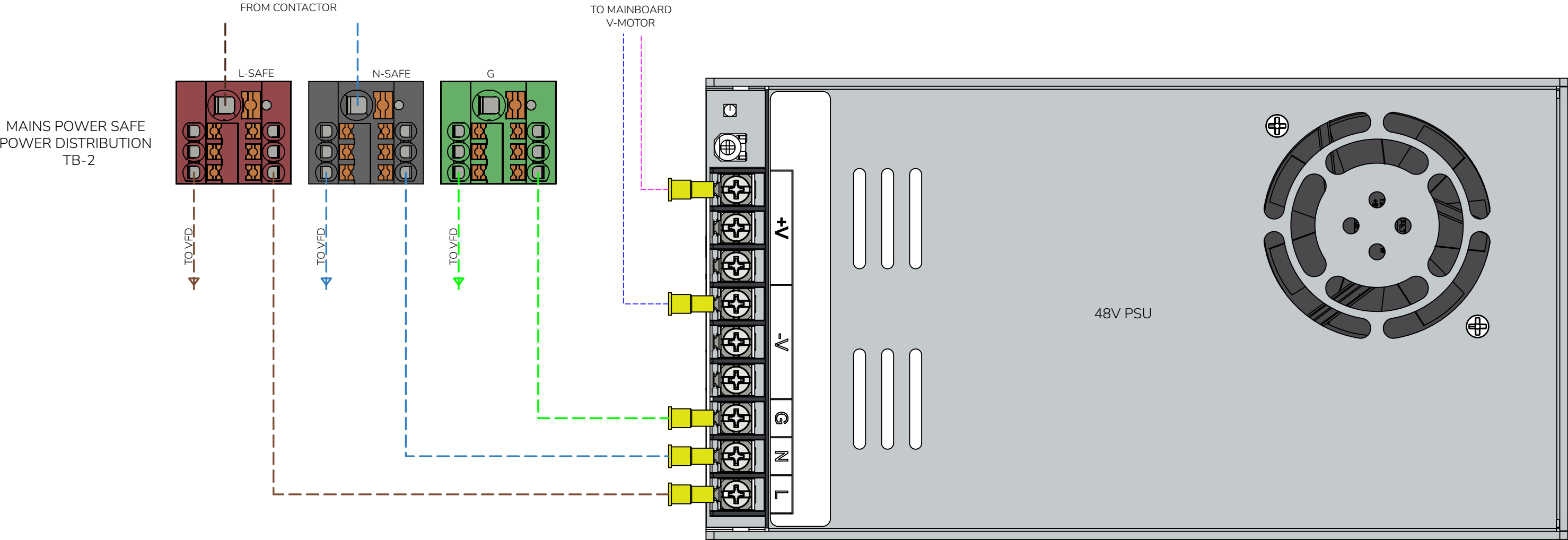
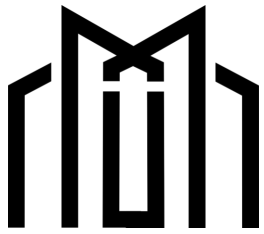
IEC INLET W/ ROCKER SWITCH COMBO

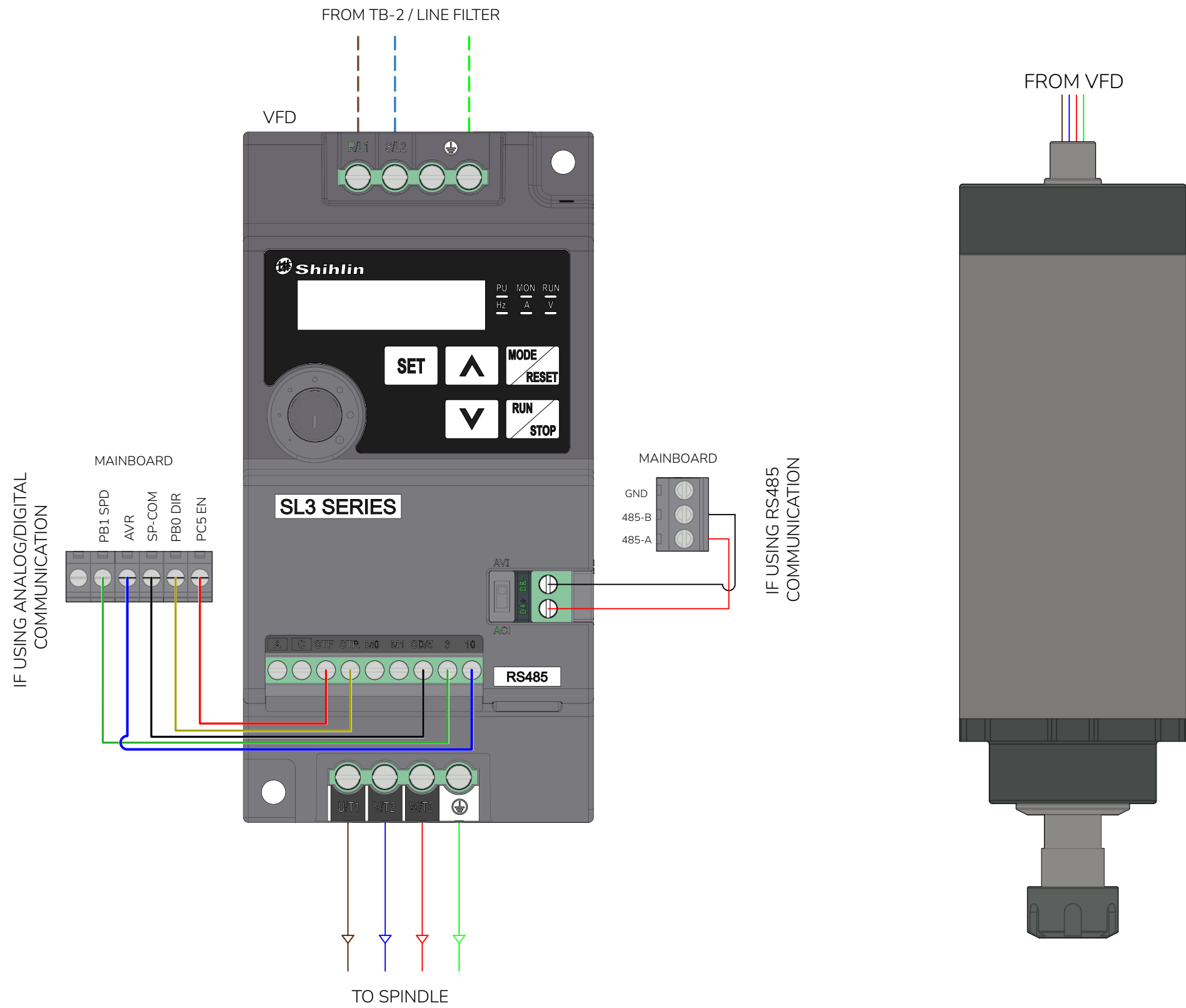


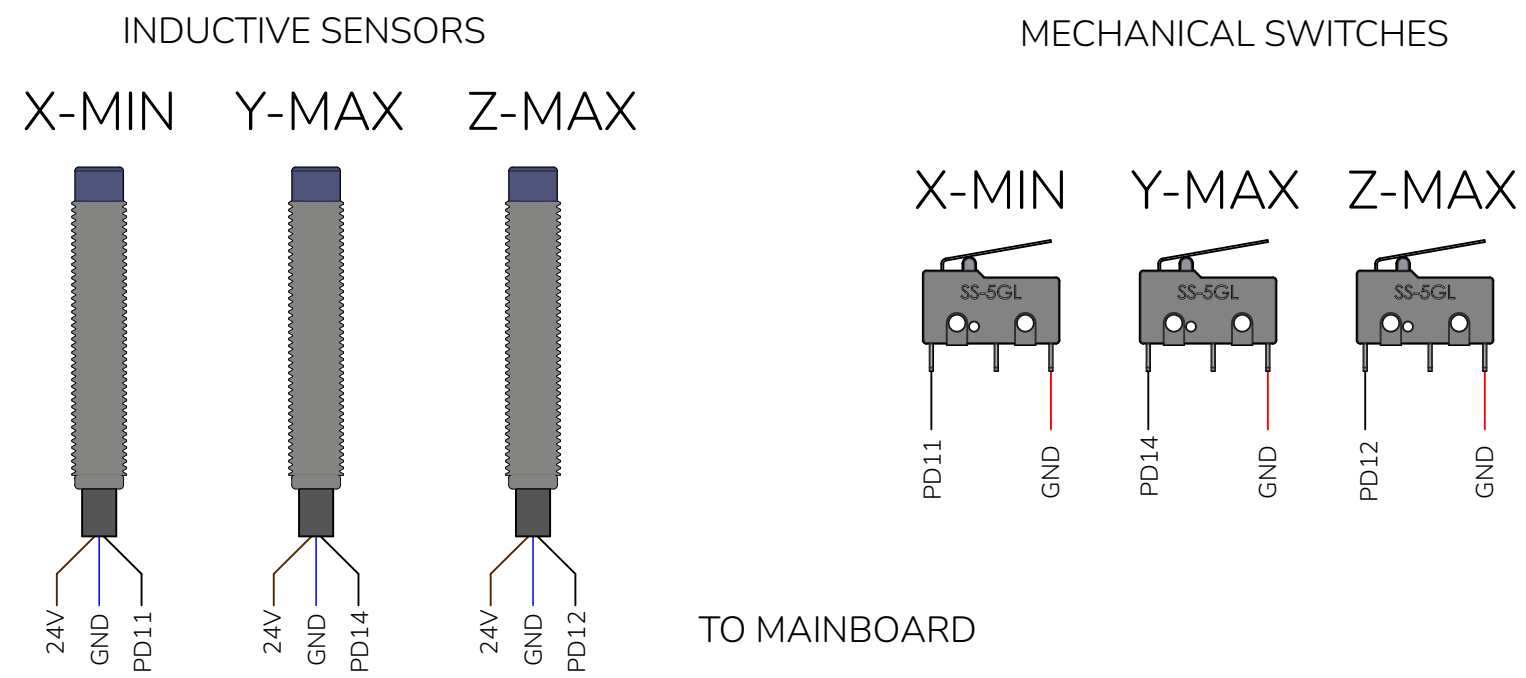
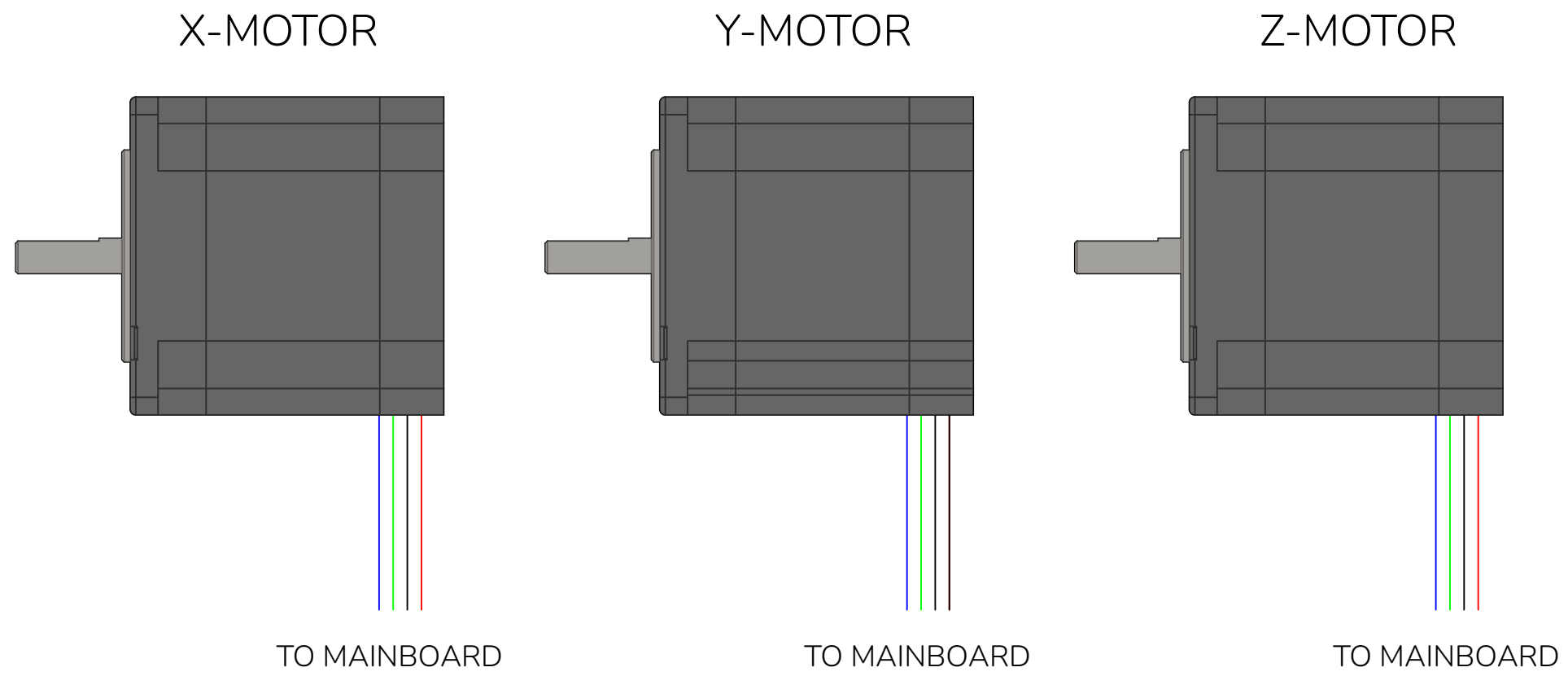
WIRING
24V / SAFE CIRCUIT

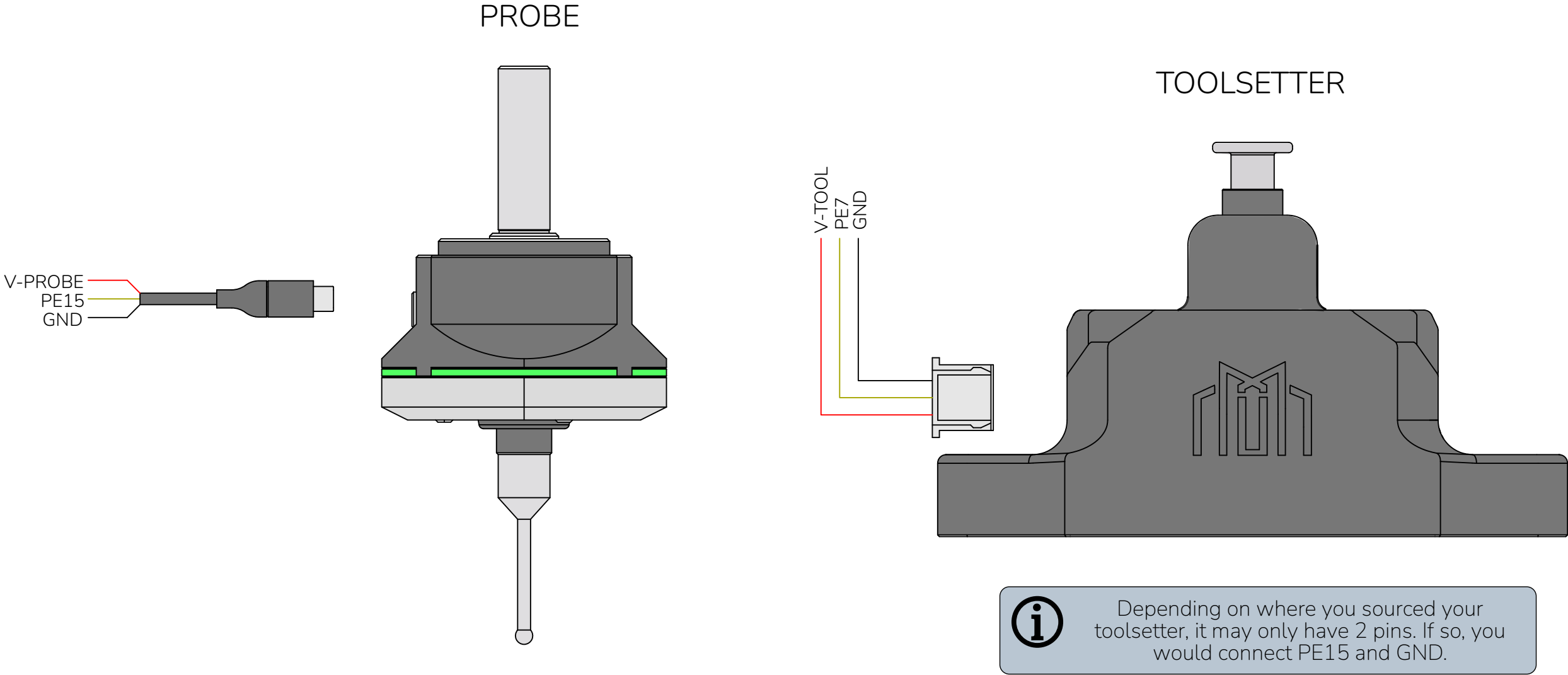
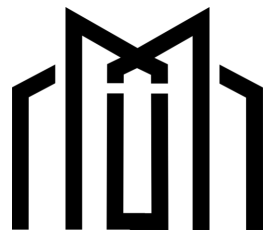


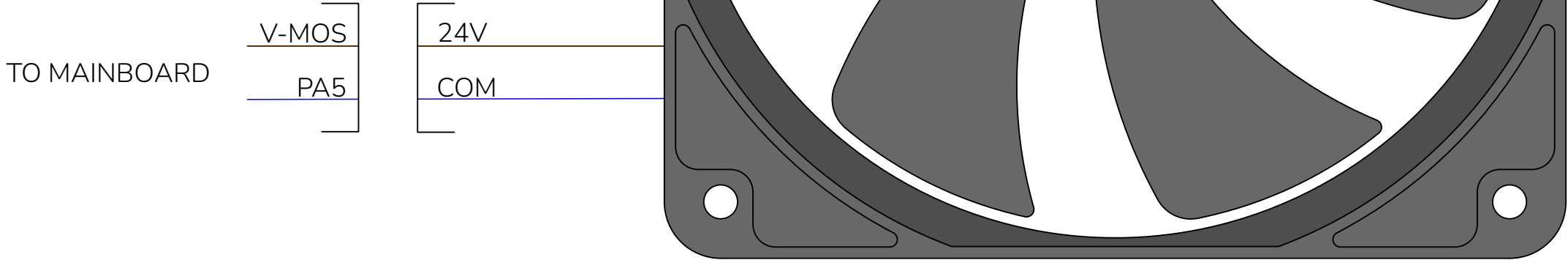
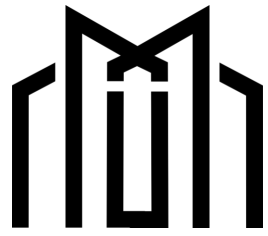


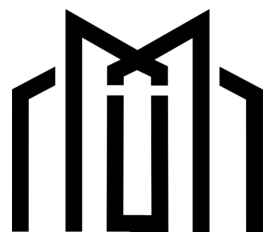




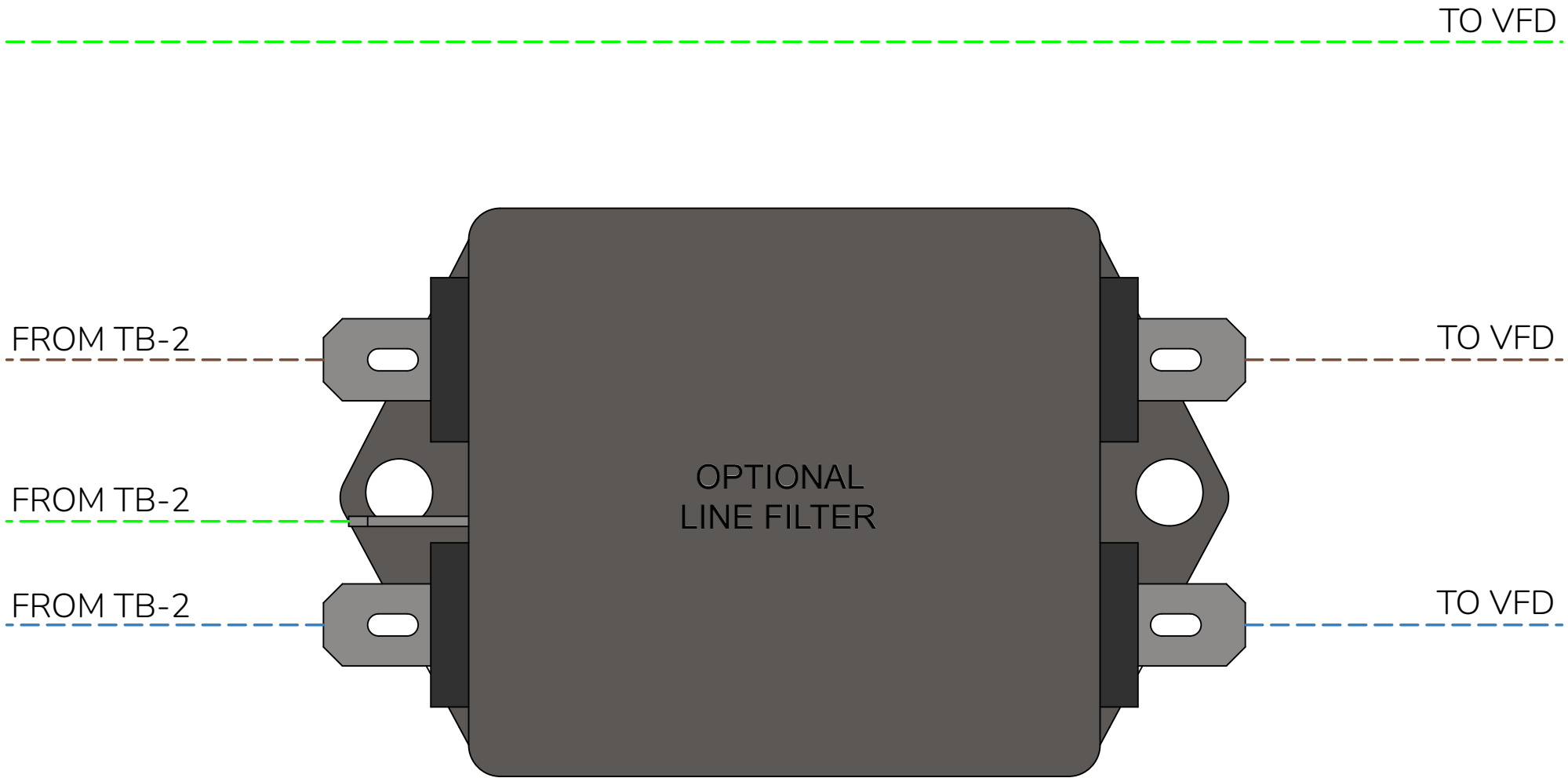


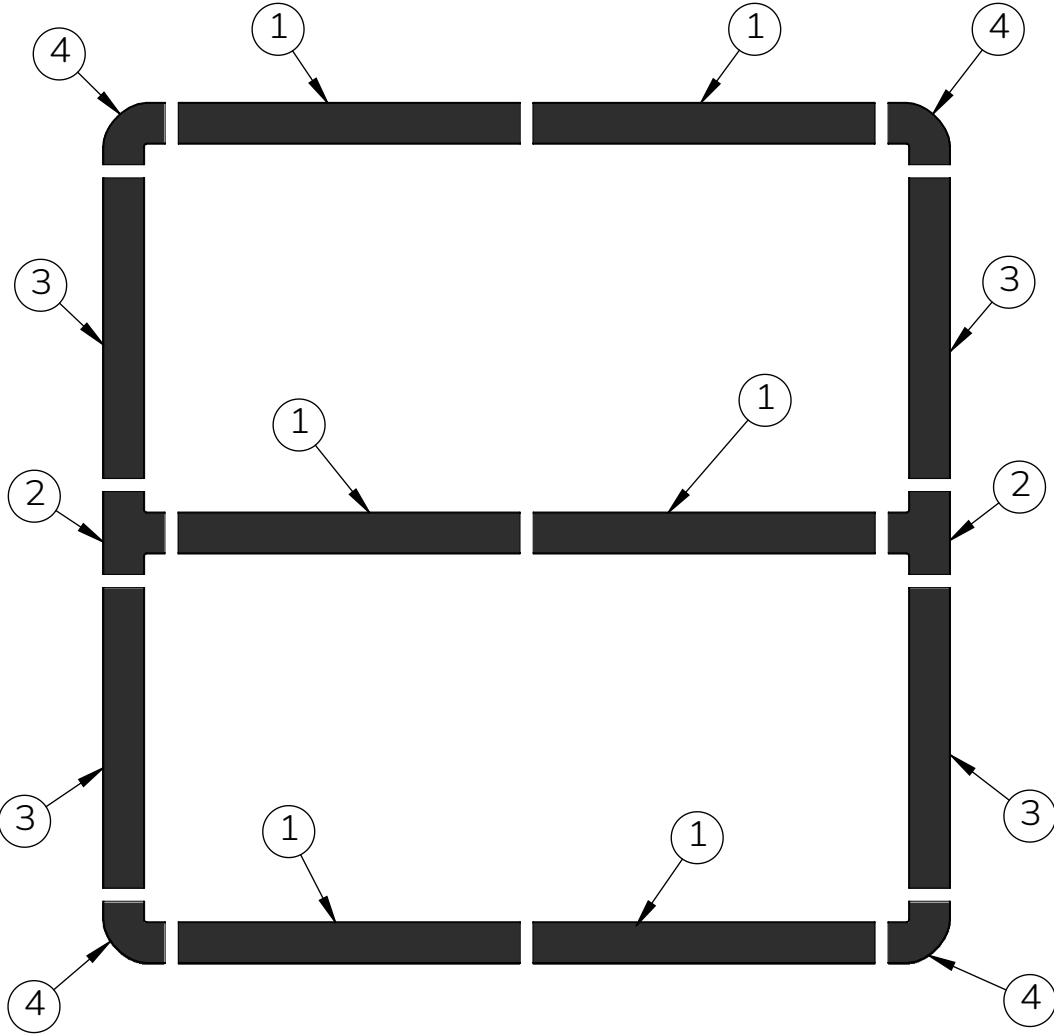
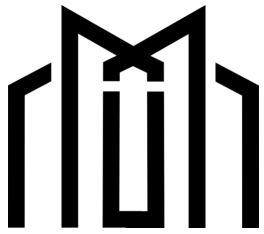




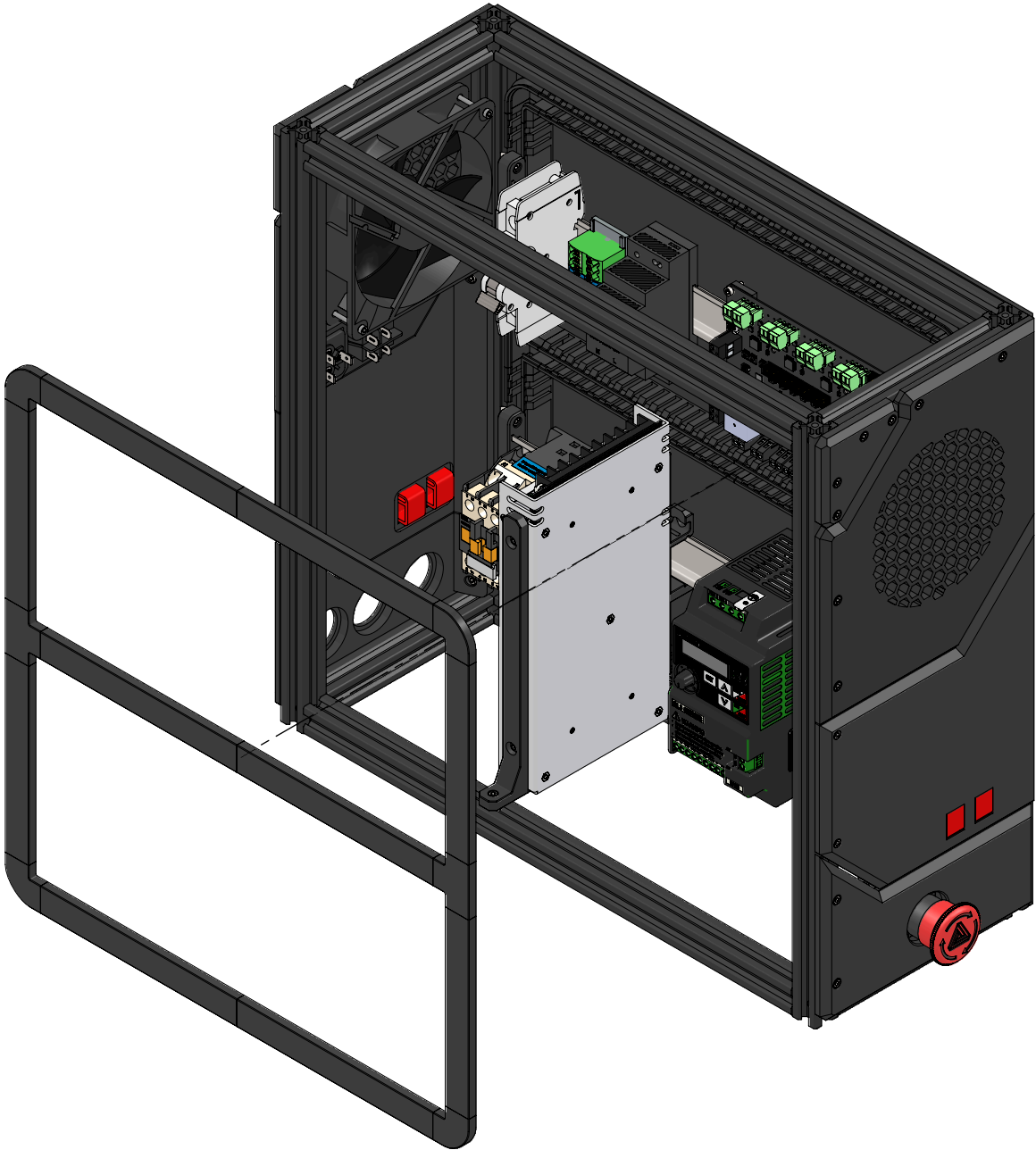


For optimal performance and reduced electrical noise, a line filter is recommended when using a spindle with a VFD. It helps minimize interference with nearby electronics and improves overall system reliability. While many users operate without issues, adding a line filter can provide extra protection against potential EMI-related problems.

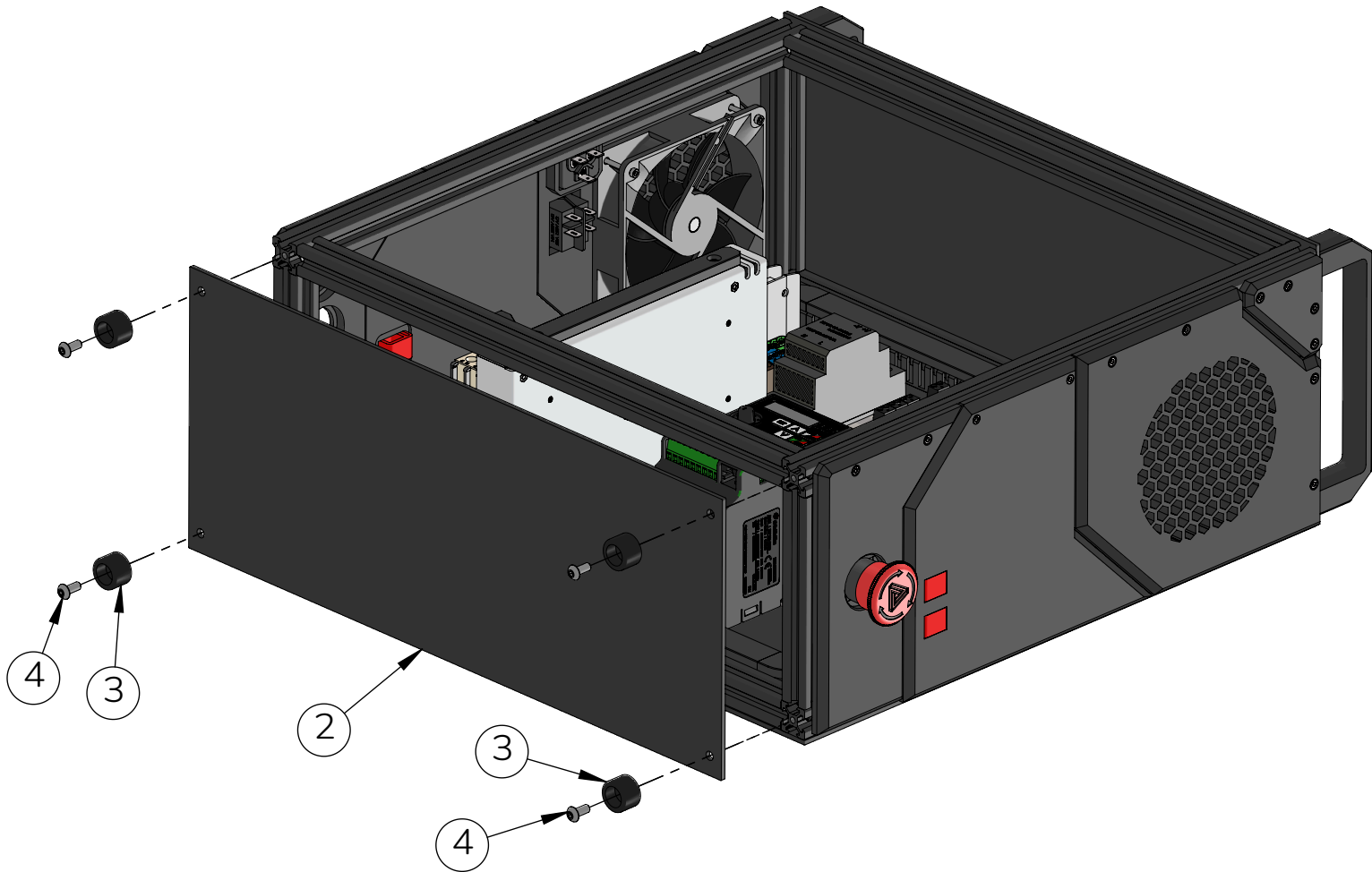
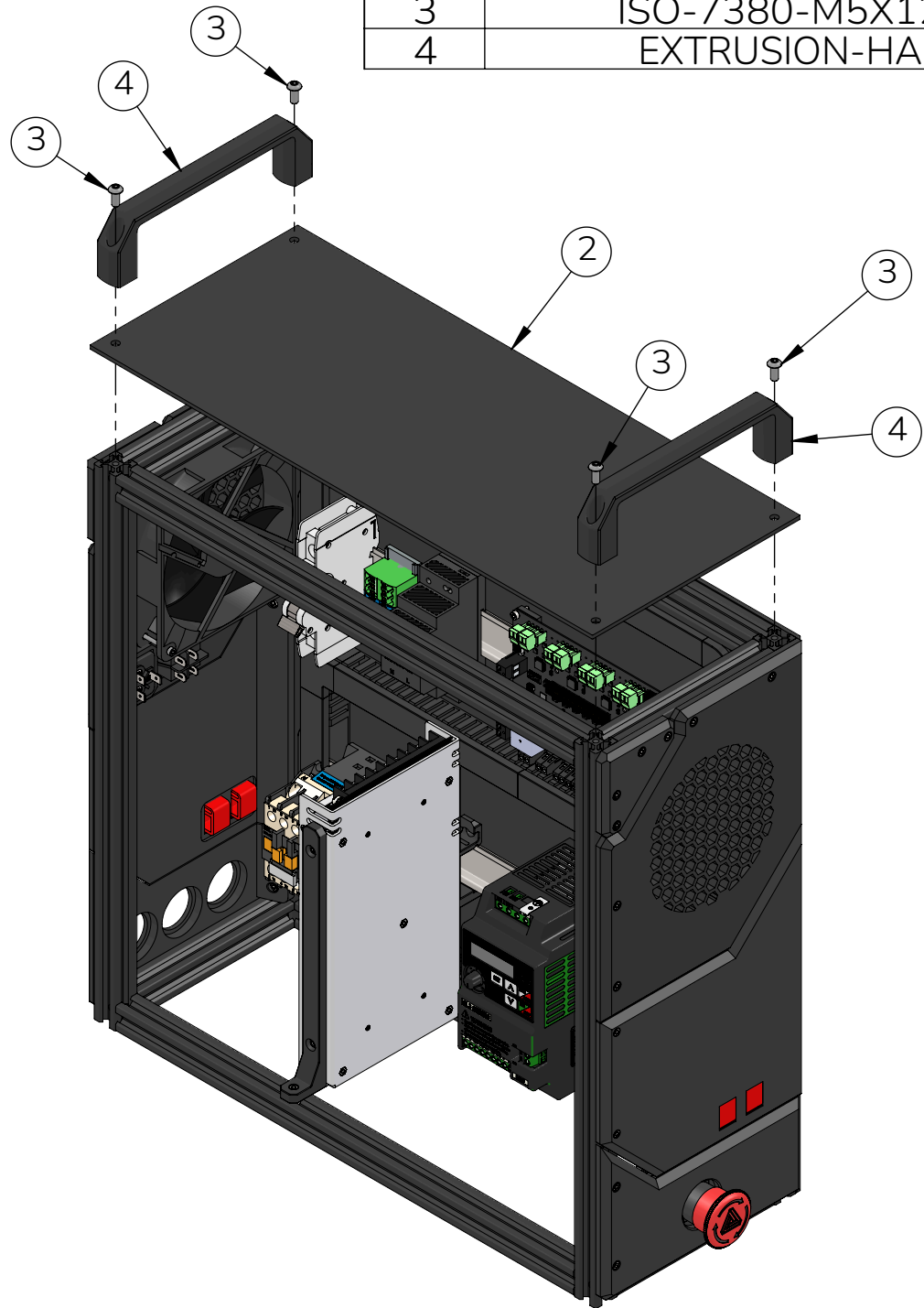




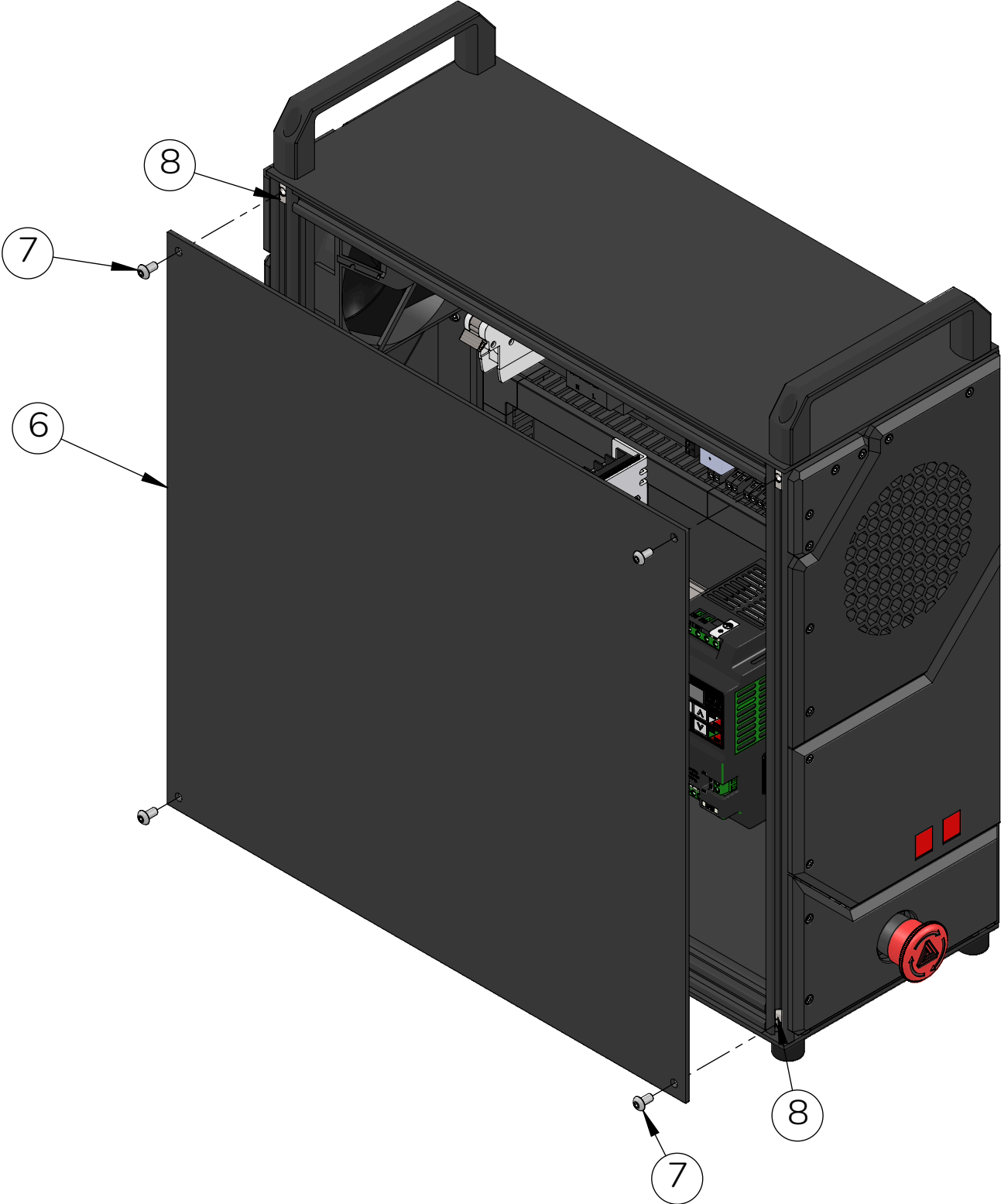
	PART NUMBER	QTY.
1	CABLE DUCT LID - 165mm	6
2	CABLE DUCT LID - T-JUNCTION	2
3	CABLE DUCT LID - 145mm	4
4	CABLE DUCT LID - ELBOW	4



	PART NUMBER	QTY.
2	450X186X3MM-PANEL	1
3	ISO-7380-M5X12 BHCS	4
4	EXTRUSION-HANDLE	2



	PART NUMBER	QTY.
2	450X186X3MM-PANEL	1
3	PRINTED-FOOT	4
4	ISO-7380-M5X12 BHCS	4



	PART NUMBER	QTY.
6	450X430X3MM-PANEL	1
7	ISO-7380-M5X10 BHCS	4
8	M5 DROP IN NUT	4